COMPUTE CHOKEPOINTS

How AI chips are made

Every stage of the supply chain, and who controls it.

The Chain: seventeen chapters, every number linked to its source.

Yashvardhan Sharma with Claude

Printed from yashvardhansharmaa.github.io/chokepoints · Updated 10 September 2026

HOW TO READ A CHAPTER

Every chapter opens with a chokepoint card. Three fields on it matter most.

- **Concentration.** How many firms credibly supply this stage: Extreme, High, Moderate or Low.
- **Chokepoint score.** 1 means many substitutes. 5 means a single point of failure, where losing one supplier stops leading-edge production worldwide.
- **Substitution time.** Our estimate of how long a well-funded state or firm would need to stand up an alternative. It is the most opinionated number on the card.

The card also gives China's position and the United States'. The chapters run in process order, and every number links to its source.

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CHAPTER
01
STAGE 1 OF 15

Chip Design, EDA and IP

The cheapest stage in the chain to fund and one of the most concentrated. Three software firms sell the tools that every leading-edge AI accelerator passes through.

IN PLAIN TERMS

Chip design turns an idea for a processor into a complete blueprint, with every one of billions of transistors placed and wired. That is the street plan of a city that size, and every road has to be walkable in less than a billionth of a second. No one draws it by hand. Software from three companies, Synopsys, Cadence and Siemens, generates the layout, checks it and signs it off, and a chip factory accepts only designs that have passed those tools. Changing one street afterwards costs a fresh set of printing stencils and several months. Withhold the software and the design cannot be made.

IN SHORT

Three vendors sell every complete design flow used at the leading edge [3]. Design is cheap next to what it commits: tens of millions of dollars of engineering and masks steer tens of billions of dollars of factory capacity [1]. It is the fastest lever a government can pull, and the one China is closest to neutralising, though CSET still puts a complete domestic flow out of reach [15]. The largest buyers now design their own parts, with Broadcom's name on the implementation rather than on the box [9].

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

5-10 years

PRICE OR MARKET SIZE

A 3 nm photomask set runs about $40M, and startups have taken a 7 nm design all the way to the factory for $50-75M

WHO LEADS

- US **Synopsys** $8B revenue in 2025 including Ansys; 90%+ of the timing-signoff market
- US **Cadence** $5.30B revenue in 2025; 55-60% of the hardware emulator market
- DE **Siemens EDA** $2.2-2.5B revenue in 2025; 85%+ of physical verification, the final layout check
- GB **Arm** About 50% of processor compute at the top cloud firms, fiscal 2026
- US **Broadcom** $16.7B of AI chip revenue in the third quarter of fiscal 2026

WHERE IT IS MADE

- US **United States** Synopsys, Cadence, Broadcom, Marvell, Nvidia, AMD; Siemens EDA's main sites
- GB **United Kingdom** Arm processor and interconnect designs, licensed from Cambridge
- TW **Taiwan** Alchip, Global Unichip and MediaTek: design services that carry a chip to the factory
- IL **Israel** Annapurna Labs, the AWS design house behind Trainium and Graviton
- IN **India** Nearly 20% of the world's chip design engineers, on the Indian government's count

WHY SUBSTITUTION IS SLOW

Swapping one tool is feasible. Replacing a whole certified flow means rebuilding the foundry-approved recipes, the timing and electrical models behind them, and twenty years of accumulated fixes for rare failure cases.

WHERE CHINA STANDS

Designs competitive accelerators, but cannot yet run a complete leading-edge flow on domestic design software, and CSET at Georgetown University judges that Chinese vendors cannot

support designs at those nodes. Washington cut off access to US and German tools for five weeks in 2025, then restored it.

WHERE THE US STANDS

Owns the two largest design-software vendors and the dominant custom-chip design houses, and controls the US-origin technology inside Siemens EDA through export licences.

Design is the only stage in this chain with no factory, and the one a foreign government can shut off fastest. In 2025 one did: for five weeks, every Chinese chip designer lost the software that turns a design into something a factory can print.

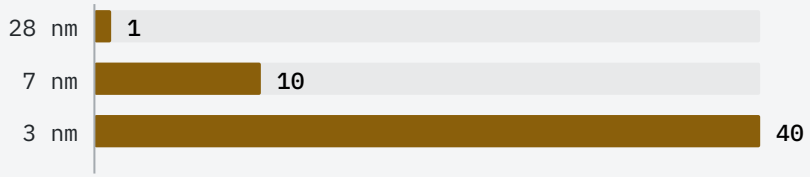
How it works

The sequence is the same at every design house:

- **Architecture.** Engineers fix the machine's shape: how wide the matrix-multiply unit is, how much fast on-chip memory sits beside it, how many high-bandwidth memory stacks line the edges.
- **RTL.** Register-transfer-level code, written in the hardware language SystemVerilog, fixes what every register holds on every tick of the clock.
- **Verification.** The biggest team on most projects: simulation, formal proofs, and emulators that boot the firmware before any silicon exists.
- **Physical design.** Software maps the code onto the foundry's prebuilt logic cells, then places them, wires them together and proves the timing holds.
- **Tape-out.** The handover. Mask data preparation and optical proximity correction, which pre-distorts each shape so it prints true, turn the layout into pattern files for the mask shop.

The discipline comes from the cost of a mistake, which is a whole new mask set. One set passes $1 million at 28 nm, $10 million at 7 nm and about $40 million at 3 nm, and a whole leading-edge chip has gone from design to tape-out on TSMC 7 nm for $50 million to $75 million, everything included [1].

What one photomask set costs, by node ($M). The 28 nm and 7 nm figures are minimums $M



Source: SemiAnalysis, The Dark Side of the Semiconductor Design Renaissance, July 2022

Variants and trade-offs

Merchant GPU versus custom ASIC

Nvidia designs Blackwell once and sells it to everyone, so the buyer gets mature software and a resale market. A custom accelerator, an ASIC built for one company's own workload, is narrower and tuned to the job in hand. Google's seventh-generation tensor processing unit, Ironwood, carries 192 GB of memory per chip, six times its predecessor's, and delivers twice the performance per watt [2].

The EDA flow

Three complete flows exist, and they do not mix. A team commits to one vendor's chain because the foundry certifies its reference recipes against specific tool versions, and timing signed off in one tool is not accepted in another. Inside each step the grip is tighter still: Synopsys alone holds more than 90% of static timing analysis, the check that proves a chip will run at its rated speed [3].

Each leading vendor's share of its own design-software segment, 2025 %



Source: SemiAnalysis, EDA Market Primer, May 2026

Buy the IP or build it

No one writes an accelerator from scratch. Design houses license the processor cores, the interface controllers and the blocks that drive high-bandwidth memory. Arm supplies most of the processor side: $4.92 billion of revenue, $2.61 billion of it royalties, and roughly 50% of processor compute at the top cloud firms in fiscal 2026 [4]. The hardest block to buy is the memory interface, which has to hold its timing across thousands of microscopic solder bumps into a stacked memory die. RISC-V, the open instruction set that needs no licence, is the housekeeping core, and Nvidia now uses it in every GPU [5].

Chiplets

The reticle limit, the largest area a scanner can print in one exposure, ended the single-die accelerator. Modern parts are chiplets, several smaller dies wired together, and the UCIe standard lets those links come from different vendors: version 3.0 adds 48 and 64 GT/s data rates [6]. Broadcom's 3.5D XDSiP goes furthest, packing more than 6,000 mm2 of silicon and 12 high-bandwidth memory stacks into one package [7].

Who makes it

Design software and licensed blocks were an $18 billion market in 2025, and Synopsys, Cadence and Siemens EDA took more than 85% of it, with no other vendor above 5% in any core category [3].

Design software and licensed-block revenue share, 2025. Synopsys includes Ansys %



Synopsys **44%** Cadence **29%** Siemens EDA **13%** All others **14%**

Source: SemiAnalysis, EDA Market Primer, May 2026

Design services, where one firm turns another's specification into a manufacturable chip, are narrower again, and Broadcom's name is on most of the deals. It booked $16.7 billion of AI semiconductor revenue in the third quarter of fiscal 2026, up 221% [8].

Its customers no longer hide the arrangement. In October 2025 OpenAI and Broadcom said they would deploy 10 gigawatts of OpenAI-designed accelerators, with Broadcom turning the design into silicon and supplying the Ethernet networking around it [9]. In April 2026 Broadcom extended its Meta partnership

to multiple gigawatts of MTIA, Meta's in-house chip, and the first 2 nm AI accelerator [10].

Below Broadcom and Marvell sit the Taiwanese houses that carry a customer's circuit list all the way to a finished mask set on TSMC's newest nodes; Alchip reported $992 million of revenue in 2025 [11]. The engineers are spread wider than the firms: India alone holds nearly 20% of the world's chip designers [12].

The chokepoint

In late May 2025 the US Bureau of Industry and Security, the agency that writes export licences, told the design-software vendors to stop selling to China. Cadence got its letter on 23 May: exporting design software classified 3D991 and 3E991 to any party in China, or to a Chinese military end user anywhere, now needed a licence [13]. Synopsys got its letter on 29 May and a rescinding letter on 2 July, after trade talks in which Washington eased technology restrictions [14].

Washington can pull this lever alone, and at once. It also costs something: it speeds up the substitution it was meant to prevent, and Synopsys's China revenue fell from 16% of the total in fiscal 2024 to 12% in fiscal 2025 [3].

China's answer is real but partial. CSET at Georgetown University puts it flatly: Chinese designers rely on US tools for all their chip designs, US firms alone sell the full range of software a leading-edge chip needs, and Chinese vendors cannot yet support designs at those nodes [15]. No domestic vendor has a certified, complete flow for a 5 nm-class accelerator, and the gap is in the rare failure cases a foundry has already validated rather than in features.

China can architect an accelerator. It cannot finish one at the leading edge without foreign tools, or build what it designs without foreign equipment: SMIC's 7 nm capacity is tight enough that Huawei has to choose between its phone chips and its AI chips [16].

Key evaluation criteria

- **Flow completeness.** How many steps one vendor covers, from the first line of code to final signoff.
- **Foundry certification.** Whether the foundry has validated that exact tool version against its reference flow and process design kit, the file set describing what the factory can print. Without that, it refuses signoff.

- **Verification throughput.** Simulation cycles and emulator capacity per day, which set the schedule more than anything else.
- **IP availability at node.** Whether the licensed blocks, the processor cores, the high-speed serial links, PCIe and the memory interface, have been proven in silicon on that process.
- **Design services depth.** Whether a partner can carry a customer's circuit list to a finished mask set on the newest node.
- **Legal exposure.** How much of the flow is US-origin technology, and so subject to export licensing.

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CHAPTER
02
STAGE 2 OF 15

Silicon and Wafers

China makes almost all of the world's polysilicon and almost none of the wafers that matter. Five firms in Japan, Taiwan, Germany and South Korea hold the 300 mm wafer every AI accelerator starts on.

IN PLAIN TERMS

Every chip starts as a wafer: a mirror-polished disc of silicon 300 millimetres across. A furnace melts purified sand, and a machine pulls a cylinder out of the melt the way a candle is dipped, except that what comes out is a single crystal, every atom on the same repeating grid. Saws slice that cylinder into discs, polished until nothing on the surface stands more than a few atoms proud. Fewer than one atom in a hundred billion may be anything but silicon, and the flatness has to hold right to the edge. Only five firms can do it reliably, and none of them is American or Chinese.

IN SHORT

The wafer is the cheapest critical input in the AI chain and among the most concentrated: five firms and $11.4 billion of revenue sit under every leading-edge chip [5] [1]. China's polysilicon dominance is real but in the wrong grade, and its 300 mm position is small [6]. The epitaxial mix matters more than the square inches, because that is where AI demand lands and where the incumbents make money. The failed move to 450 mm showed that this layer changes over decades.

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

High

SUBSTITUTION TIME

5-10 years

PRICE OR MARKET SIZE

The entire world market was $11.4bn of wafer revenue on 12,973 million square inches shipped in 2025

WHO LEADS

- JP **Shin-Etsu Handotai** Largest supplier; the parent's electronics materials arm sold ¥750.3bn from April to December 2025
- JP **SUMCO** ¥409.7bn of 2025 net sales, but an ¥11.8bn loss
- TW **GlobalWafers** NT$60.6bn revenue in 2025
- KR **SK Siltron** One of the five firms behind the whole wafer market
- DE **Siltronic** €1,346.7m revenue in 2025

WHERE IT IS MADE

- JP **Japan** Shin-Etsu and SUMCO crystal growth and polishing
- TW **Taiwan** GlobalWafers headquarters and plants
- DE **Germany** Siltronic Burghausen and Freiberg; Wacker semiconductor-grade polysilicon
- KR **South Korea** SK Siltron, being sold to Doosan
- US **United States** Hemlock polysilicon in Michigan; GlobalWafers Sherman, Texas and St Peters, Missouri

WHY SUBSTITUTION IS SLOW

Crystal pulling and final polish rest on know-how that was never written down, and each new wafer line has to be re-qualified for every chip at every fab before one production wafer ships.

WHERE CHINA STANDS

Made about 93% of the world's polysilicon in 2023, but 98% of that was solar-grade, too impure for chips. Held under 1% of the 300 mm wafer market as of 2021, and supplies only 12% of the wafers its own 300 mm fabs consume.

WHERE THE US STANDS

No US-headquartered volume wafer maker. Hemlock supplies polysilicon from Michigan; GlobalWafers took a CHIPS award of up to $406m for a $3.5bn Texas plant that opened in May 2025.

The whole world market for silicon wafers came to $11.4 billion of revenue on 12,973 million square inches in 2025 [1]. That is a small fraction of Nvidia's data center revenue, and every leading-edge chip behind that revenue was built on one of these discs.

How it works

Silicon starts as quartzite, which an arc furnace strips of oxygen with carbon. Refiners then turn that into trichlorosilane gas, distil it, and decompose it onto heated filaments inside a bell jar. This Siemens process grows the grey polysilicon rods everything downstream is made from. The same process serves solar cells and chips, but only the ultra-pure semiconductor grade is any use here [2], and the two grades are separate industries with separate suppliers.

A grower melts the polysilicon at about 1,420 degrees Celsius and pulls a single crystal out of it by the Czochralski method, starting with a thin neck that chokes off the flaws formed at the seed. Boron or phosphorus stirred into the melt, a step called doping, sets how readily the wafer conducts and whether its charge carriers are positive or negative. A wire saw slices the ingot, then the maker laps, etches and mirror-polishes each slice with colloidal silica. Epitaxial wafers get one more layer of near-perfect silicon grown on top at about 1,200 degrees Celsius [3], and almost every step throws material away.

Variants and trade-offs

Prime versus epitaxial

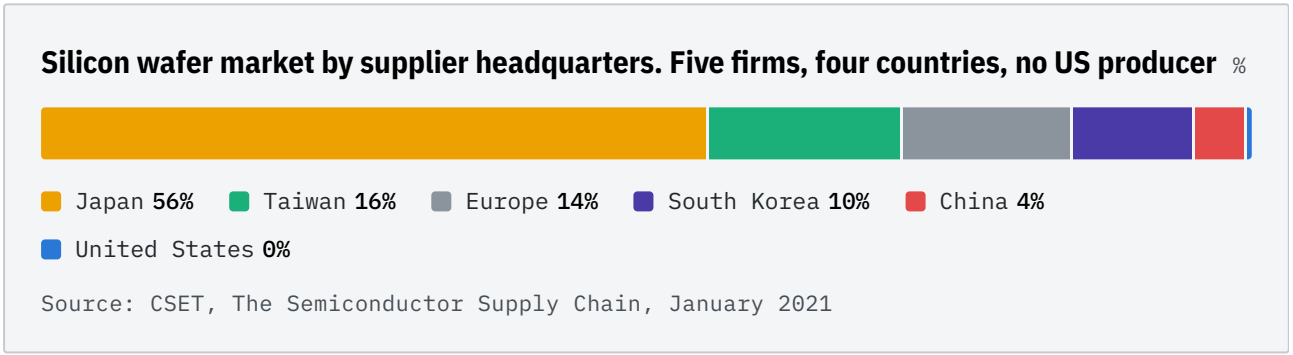
A prime wafer is a uniformly doped, front-polished single crystal: cheapest, and adequate for most memory and mature logic. An epitaxial wafer adds a thin, lightly doped, defect-controlled layer on top. The transistors live in that layer, so its quality sets the floor on leakage and on latch-up, a parasitic short between power and ground. SEMI, the industry association, credits the 2025 rise in shipments to AI demand for advanced epitaxial wafers and high-bandwidth memory [1].

Silicon on insulator

A buried layer of oxide, which is glass, separates a thin working layer of silicon from the bulk beneath, and Soitec's Smart Cut process makes nearly all of it [4]. It matters for radio and low-power chips, not for AI GPUs, which run on plain epitaxial silicon.

Who makes it

Siltronic's own map of the chain puts five major suppliers behind the whole market [5]. CSET at Georgetown names them: Shin-Etsu, SUMCO, GlobalWafers, Siltronic and SK Siltron, headquartered in Japan, Taiwan, Germany and South Korea [6].



Their 2025 results split between an AI boom and a slump in the older, mature nodes.

- **Shin-Etsu.** Its electronics materials arm sold ¥750.3 billion in the nine months to December 2025, on strong AI-related wafer demand [7].
- **SUMCO.** Sold ¥409.7 billion and still lost ¥11.8 billion, as operating profit fell from ¥36.9 billion to ¥1.3 billion while it added leading-edge 300 mm capacity and reorganised its weak 200 mm lines [8].
- **Siltronic.** Lost €77.9 million on €1,346.7 million of sales while spending €369.1 million on its Singapore fab [9].
- **GlobalWafers.** NT$60.6 billion of revenue, down 3.24 percent in local currency [10].

Price explains the gap: area shipped rose 5.8 percent in 2025 while revenue fell 1.2 percent [1]. Wafers sell on multi-year agreements that fix price and volume years ahead, so a demand spike never reaches the invoice.

The raw material is a separate map. China made 1.50 million tonnes of polysilicon in 2023, about 93 percent of world output [11]. Almost none of it can

go into a chip: solar-grade material was 98 percent of that output and electronic-grade 2 percent [11].

China's polysilicon output by grade, 2023. China made about 93% of the world's 1.5 million tonnes that year %



Solar-grade **98%** Electronic-grade **2%**

Source: USGS Minerals Yearbook, China, 2023

Siltronic sizes electronic-grade silicon at $1.4 billion, with five major suppliers, none of them Chinese: Wacker, Hemlock, OCI, Tokuyama and Mitsubishi [5].

Crystal growth and polishing sit in Japan, Taiwan, Germany and South Korea, with no major US-headquartered producer as of 2021 [6]. That is starting to change. GlobalWafers opened a $3.5 billion plant in Sherman, Texas in May 2025, the first production line of its kind built in the United States in over twenty years, and announced another $4 billion for phases three and four the same day [12]. Washington had awarded it up to $406 million under the CHIPS incentives programme [13].

The chokepoint

China dominates polysilicon and barely registers in wafers. As of 2021 Chinese firms held under 5 percent of the wafer market and under 1 percent of 300 mm [6]. They cannot even feed their own fabs: Chinese producers supplied 12 percent of the wafers consumed by China's 300 mm fabs, against 18 percent at 200 mm and below [6]. That is the base domestic makers are building from.

Leading-edge 300 mm wafers rest on know-how that was never written down rather than on any machine a rival could buy, and 300 mm carried 99.7 percent of world fab capacity at 45 nm and below as of 2021 [6]. A fab has to qualify each new wafer source for every chip it makes, which takes quarters, while the contracts already in place run for years.

The last increase in wafer size, to 300 mm, happened in 2002 [6]. Bigger discs make printing a bigger share of the bill: photolithography takes half the cost of processing a 300 mm wafer against a quarter of a 150 mm one [6]. The push to 450 mm never reached production, and every leading-edge chip is still made on 300 mm.

AI accelerators make the wafer's job harder. One fatal defect ruins far more silicon on a GPU die that fills a whole printed field than on a small chip, and extreme-ultraviolet printing has almost no depth of focus to give (see Lithography), so flatness has to hold everywhere on the disc rather than on average.

Key evaluation criteria

- **Grade.** Electronic against solar polysilicon is the line between a Chinese commodity market and a German, American, Japanese and Korean one; solar was 98 percent of China's 2023 output [11].
- **Defect density** in the epitaxial layer. A big AI die loses more good silicon per defect than a small chip does, which is why AI demand shows up first in advanced epitaxial wafers [1].
- **Flatness and nanotopography**, the fine ripple across the surface, because extreme-ultraviolet exposure has almost no depth of focus to spare across a 300 mm wafer.
- **Diameter**, where 300 mm carries every leading-edge node and 99.7 percent of capacity at 45 nm and below [6].
- **Contract structure**, since long-term agreements set price and volume years ahead and blunt a demand spike.

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CHAPTER
03
STAGE 3 OF 15

Chemicals, Gases and Photoresist

A few hundred molecules stand between a blank wafer and a working chip. Japan makes about 90% of the photoresist, and the two times this layer has been squeezed, no fab stopped: the countries hit stockpiled and built substitutes instead.

IN PLAIN TERMS

Photoresist is photographic film for chips. A coating machine spreads a film thinner than a soap bubble across the wafer, light shines through a stencil, and the exposed parts dissolve away, leaving a pattern the next machine can etch. A few hundred other gases and liquids deposit, strip and clean around it. All of it has to be clean to about one stray metal atom in a trillion, and each formula is approved for one factory and one product at a time. That is why nine tenths of the world's photoresist comes from Japan, and a second source takes years rather than weeks.

IN SHORT

Photoresist is the sharpest point here: about 90 percent Japanese [3], with one of the largest suppliers now 84 percent owned by a Japanese state fund [4]. Individual molecules are far more concentrated than the industry selling them: neon and nitrogen trifluoride come from a handful of plants. China's gallium, germanium and antimony controls bite on non-silicon chips and optics but not on a silicon GPU, and the gallium suspension runs for one year from November 2025 [13]. The qualification cycle matters more than the spot price, because every disruption so far was absorbed within a year and each one added a second source somewhere.

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

High

SUBSTITUTION TIME

3-7 years

PRICE OR MARKET SIZE

Wafer fabrication materials were a $45.8bn market in 2025 and packaging materials $27.4bn, $73.2bn between them

WHO LEADS

- JP **JSR** Owner of Inpria's metal-oxide resist for extreme-ultraviolet printing; 84% held by Japan Investment Corporation, a state-backed fund, since April 2024
- JP **Tokyo Ohka Kogyo** Top-five resist maker; ¥139.7bn of sales in the first half of fiscal 2026
- FR **Air Liquide** €2,465m of electronics revenue in 2025, 9.1% of group sales
- US **Entegris** $3.20bn of 2025 revenue in filters, polishing slurries, gas delivery and wafer carriers
- JP **JX Advanced Metals** About 65% of sputtering targets, the metal slugs that supply chip wiring

WHERE IT IS MADE

- JP **Japan** Photoresist, ultra-pure hydrofluoric acid, sputtering targets, polishing slurries
- US **United States** Entegris filters and wafer carriers, DuPont polishing slurries, Air Products gases
- FR **France** Air Liquide electronics gases and precursors
- KR **South Korea** SK Materials nitrogen trifluoride, Dongjin resist, on-site gas for memory fabs
- CN **China** Gallium, germanium and antimony refining; growing wet chemicals and polishing slurries

WHY SUBSTITUTION IS SLOW

Every formula is qualified for one process at one fab, and metal impurities are counted in parts per trillion, so a second source takes years of joint development rather than a purchase order.

WHERE CHINA STANDS

Cannot make resist for extreme-ultraviolet or leading-edge 193 nm printing, and held under 5% of the photoresist market as of 2021. It refines 99% of the world's primary low-purity gallium and has licensed exports of gallium and germanium since August 2023.

WHERE THE US STANDS

Strong in filters, containers, polishing slurries and gas delivery through Entegris, DuPont and Air Products. Nearly absent from advanced photoresist since JSR bought Inpria.

A modern fab is a chemical plant with lithography attached. Air Liquide alone supplies more than 200 molecules and roughly 50,000 cylinders a year [1]. None of it costs much by the tonne, and any one of them missing stops the line.

How it works

Since the 1990s the workhorse has been the chemically amplified resist: one photon releases one acid molecule, which then drives hundreds of reactions in a bake after exposure. That lets a scanner expose a wafer in milliseconds, at 248 nm from a krypton fluoride laser or 193 nm from an argon fluoride laser shining through water.

Extreme ultraviolet breaks that bargain (see Lithography). A 13.5 nm photon carries about fourteen times the energy of a 193 nm one, so the same dose of light arrives as far fewer photons, and the luck of where each one lands shows up as ragged line edges and missing contact holes. Resist makers answered by absorbing more of the light: metal-oxide resists, built on clusters of tin and oxygen, take up extreme ultraviolet far better than carbon films, which is why JSR bought Inpria in 2021 [2].

Everything else puts material on the wafer, takes it off, or keeps it clean. Bulk gases arrive by the tonne; specialty gases in grams, from silane for deposition to nitrogen trifluoride, split apart in a plasma to strip the chamber walls. Hydrofluoric acid cleans, polishing slurries flatten each layer, sputtering targets supply the metal for the wiring, and filters and carriers keep contamination out.

Variants and trade-offs

RESIST CLASS	WAVELENGTH	WHERE IT IS USED	LIMIT
i-line novolak	365 nm	Packaging, mature nodes	Resolution
KrF chemically amplified	248 nm	Implant layers, thick films	Resolution
ArF immersion chemically amplified	193 nm	Most critical layers below 28 nm	Needs several exposures per layer
EUV chemically amplified	13.5 nm	Leading-edge single-exposure layers	Shot noise from too few photons
EUV metal-oxide	13.5 nm	High-NA tools, tightest pitches	Cost, defects, outgassing

Resist classes are not substitutes; each is tied to a scanner and a layer [3].

Who makes it

Japan produced about 90 percent of the world's semiconductor photoresist as of 2021, the rest mostly in the United States and South Korea [3].

Semiconductor photoresist production share by country, 2021 %



Japan **90%** United States, South Korea and others **10%**

Source: CSET, The Semiconductor Supply Chain

JSR is no longer a public company. A vehicle of Japan Investment Corporation, a state-backed fund, took 84.36 percent of its shares at ¥4,350 each in a tender offer that closed on 16 April 2024, then delisted it [4].

Tokyo Ohka Kogyo sold ¥139,668 million in the first half of fiscal 2026, up 25.1 percent on AI server demand [5]. Fujifilm lifted its polishing-slurry capacity at Kumamoto by 30 percent in January 2025 [6] and opened extreme-ultraviolet resist lines at Shizuoka and Pyeongtaek that October [7].

The layer is bigger than it looks: wafer fabrication materials sold $45.8 billion in 2025, packaging materials $27.4 billion [8].

Gases are a Western-led oligopoly: Linde, Air Liquide, Air Products, Nippon Sanso and Messer. Air Liquide's electronics arm alone sold €2,465 million in 2025, 9.1 percent of group revenue [1]. Under them sit single-molecule specialists: SK Materials in nitrogen trifluoride, Kanto Denka in fluorine chemistry, Stella Chemifa and Morita in ultra-high-purity hydrofluoric acid.

The consumables are narrower again. As of 2021 five firms held more than 60 percent of wet chemicals, and DuPont and Cabot Microelectronics about 56 percent of a $790 million market in polishing slurries [3]. One material is tighter than all of them: JX Advanced Metals holds about 65 percent of the world sputtering-target market [9].

The chokepoint

This layer has been squeezed twice since 2019, once by a government and once by a war.

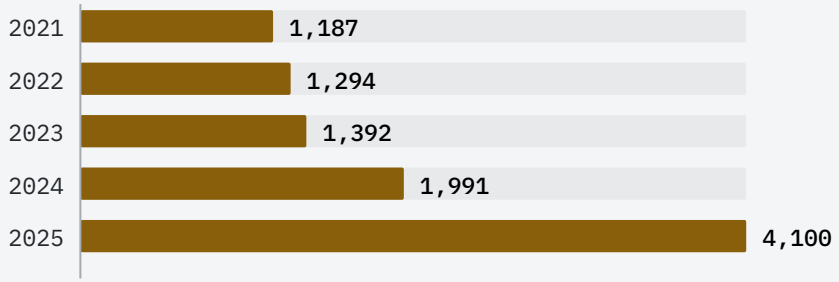
In July 2019 Japan put photoresist, hydrogen fluoride and fluorinated polyimide under individual export licences for South Korea. It had supplied more than 90 percent of Korean imports of two of the three, and hydrogen fluoride exports fell 87.9 percent [10]. Korea kept its fabs running and spent five years funding domestic substitutes, which is what materials controls usually produce.

In 2022 Russia invaded Ukraine, which supplied about 70 percent of the world's neon and 90 percent of the semiconductor-grade neon used by US industry [11]. Buyers had stockpiled since the 2014 annexation of Crimea, and Korean and Japanese chipmakers reported adequate neon from China instead [11].

China's lever is the metals rather than the process chemicals. Beijing announced export licences for gallium and germanium in July 2023, in force from August, then banned both, plus antimony and superhard materials, to the United States in December 2024 [12]. In November 2025 it lifted the gallium ban for one year [13].

Neither metal goes into a silicon logic chip. Germanium goes into fibre optics, infrared optics, solar cells for satellites and radiation detectors [14]; gallium into the non-silicon wafers behind radio front ends, LEDs and lasers [13]. What they move is price. Germanium metal averaged $4,100 a kilogram in 2025 against $1,392 in 2023, and US imports of the metal fell 67 percent [14]. The US Geological Survey put the cost to US GDP of a complete ban at about $3.4 billion [12].

Germanium metal, annual average price ($/kg). China licensed exports from August 2023 and banned US shipments in December 2024 $/KG



Source: USGS Mineral Commodity Summaries, February 2026 (Argus Media prices)

The slower risk is regulatory. The five member states behind the European Union's proposed restriction on PFAS, the long-lived fluorinated chemicals, published a revised text on 20 August 2025, after working through more than 5,600 comments [15]. The committees of ECHA, the EU chemicals agency, now back an EU-wide restriction with targeted exemptions rather than a flat ban, and the scientific evaluation is due to finish by the end of 2026 [15]. These chemicals are everywhere in a fab: photoacid generators, etch gases, chamber liners, pump seals and filter membranes are all fluorinated. We think an exemption is likely, but nothing is law yet.

Key evaluation criteria

- **Purity, in parts per trillion** of metallic contamination, not per million. This is the spec that keeps entrants out.
- **Qualification lock-in**: a resist or slurry is approved for one layer of one product at one fab, so switching costs are counted in tape-outs.
- **Photon efficiency** for EUV resists: how much light the resist absorbs sets the dose, the dose sets how many wafers an hour the scanner prints, and too few photons per dose set the defect rate. That is the case for metal-oxide chemistry.
- **Transportability**: bulk gases are made at the fab fence, and only cylinder-shipped specialty gases can be embargoed.
- **Byproduct exposure**: gallium is recovered mostly from bauxite refining and germanium from zinc concentrates, so supply answers to another industry's economics [13].

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CHAPTER
04
STAGE 4 OF 15

Lithography

The most concentrated stage in the chain. One firm in the Netherlands builds every extreme ultraviolet scanner in the world, the machine that prints the finest circuit layers. It shipped 48 of them in 2025, at around $200 million each.

IN PLAIN TERMS

Lithography prints the pattern of a circuit onto the wafer: the machine holds a stencil of one layer up to a lamp and projects it, four times smaller, onto the silicon. Finer circuits need shorter wavelengths, and the shortest light now in use will not pass through air, so the printing has to happen in a vacuum. Only one company has ever built such a machine, and it is Dutch. Export controls on advanced chipmaking run through that one firm.

IN SHORT

ASML, in Veldhoven, using Zeiss mirrors from Oberkochen, sets the resolution of every leading-edge AI chip. The tool count is small enough to track by hand: 48 EUV systems booked in 2025, four of them High-NA [1]. Export controls cut China's share of ASML system sales from 41% in 2024 to 33% in 2025 [3], and China's answer is domestic immersion DUV plus an EUV prototype that has not yet made a chip. No announced programme changes this bottleneck before 2030.

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

10+ years

PRICE OR MARKET SIZE

About $200M

for a standard EUV scanner, $350-400M for the newer High-NA version and about $60M for a DUV tool, on Reuters' reporting. ASML does not publish per-system prices

WHO LEADS

- NL **ASML** 100% of EUV (extreme ultraviolet) and above 80% of DUV (deep ultraviolet); EUV was 48% of its own 2025 system revenue and immersion DUV 42%
- JP **Nikon** 22 new chipmaking scanners in the year to March 2026, in a market Nikon sizes at 570 units
- JP **Canon** i-line and krypton fluoride steppers for older, coarser layers, plus the only commercial nanoimprint tool

WHERE IT IS MADE

- NL **Netherlands** ASML design and final assembly, Veldhoven
- DE **Germany** Zeiss SMT optics, Oberkochen; Trumpf carbon dioxide lasers that drive the light source, Ditzingen
- US **United States** ASML light-source research and manufacturing, San Diego
- JP **Japan** Nikon and Canon deep ultraviolet scanners and steppers

WHY SUBSTITUTION IS SLOW

ASML built its first EUV prototype in 2001 and needed nearly two decades and billions of euros before commercial chips shipped in 2019. It still depends on Zeiss mirrors that no one else can make.

WHERE CHINA STANDS

No EUV scanner has ever been sold to a customer in China, and CSIS judges the technology firmly out of reach despite state investment. A Shanghai firm, Aishengna, started building

domestic immersion DUV scanners in 2026. SMEE, the older domestic supplier, holds about 4% of the global i-line market, the coarsest tier, and nothing above it.

WHERE THE US STANDS

No domestic scanner maker. Washington controls access through Dutch licensing, US-origin subsystems and entity-list rules.

Every transistor in an AI accelerator gets its shape from a machine one company builds. ASML sold 327 lithography systems in 2025, 48 of them extreme ultraviolet, and nobody else sells EUV [1].

How it works

The stencil is a photomask, the film is photoresist, the machine is a scanner (see Photomasks and pellicles).

Resolution follows one relation: the finest feature a scanner can print is a process factor times its wavelength, divided by the numerical aperture, which is how wide a cone of light the optics gather. A wider cone prints finer, and a denser medium widens it, so ASML floods the gap above the wafer with ultrapure water and reaches an aperture of 1.35 on its deep ultraviolet systems [2].

When one exposure cannot resolve a pattern, the fab prints the layer with two, three or four masks in sequence. Each pass costs another mask, more time on the tool and another chance to misalign, so moving a layer to a single EUV exposure usually saves money.

Variants and trade-offs

Deep ultraviolet

Deep ultraviolet (DUV) light comes from gas lasers: 248 nanometres from krypton fluoride (KrF), 193 from argon fluoride (ArF), shone through air or water. These tools still dominate by volume: of ASML's 327 systems in 2025, 131 were ArF immersion, 78 KrF, 54 the older, coarser i-line and 16 ArF dry [3]. A DUV scanner costs roughly $60 million, the one tier with named competitors, Nikon and Canon, where ASML still holds above 80% [4].

Immersion is the contested tier. Without EUV, a fab needs NXT:2000i-class scanners and several passes per layer to reach leading-edge logic, and that is where Dutch export licensing draws its line.

Extreme ultraviolet

The 13.5 nm light comes from dropping molten tin into a vacuum vessel 50,000 times a second and hitting each droplet twice, with a low-intensity pulse to flatten it and then one that vaporises it into plasma [5]. Zeiss mirrors carry it to the wafer, and the largest are a metre across, polished flat to within tens of picometres, thousandths of a nanometre [2].

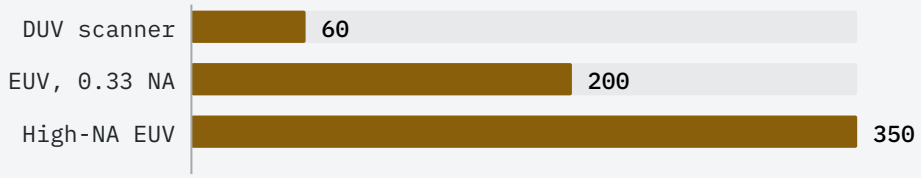
Brighter light means more wafers an hour, and ASML ran the first 1,000-watt EUV source in April 2025 [6]. Its TWINSCAN NXE:3800E now ships at its full 220 wafers an hour, 37 percent better than the NXE:3600D [6].

High-NA EUV

High numerical aperture (High-NA) raises the aperture from 0.33 to 0.55 and cuts the smallest printable feature from 13 nm to 8 nm [7]. Its optics shrink the mask image four times in one direction and eight in the other, halving the patch of wafer one exposure covers and doubling the exposures per wafer [7]. The second-generation TWINSCAN EXE:5200B runs at 175 wafers an hour, 60 percent above the EXE:5000 [6].

Reuters puts a standard EUV tool at around $200 million and a High-NA machine at $350 to $400 million, prices ASML does not publish [4].

What one lithography system costs. The High-NA figure is quoted as a $350-400M range; the bar shows the low end $M



Source: Reuters, The $400 million ASML 'printers', July 2026

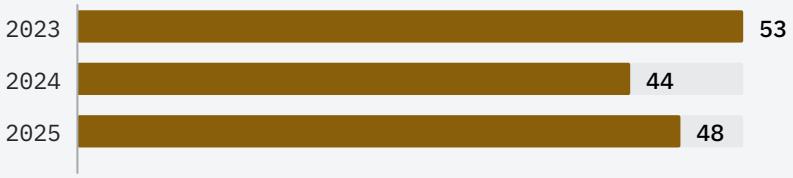
Intel and TSMC have split on High-NA:

- **Intel Foundry** ships the first high-volume logic product made with High-NA, on Intel 18A, and installed the first EXE:5200B [8], after more than a million High-NA wafers [9].

- **TSMC** held off, and now says it will use High-NA in high-volume manufacturing from 2030 [10].

ASML booked four EXE systems as revenue in 2025 against two in 2024 [1], and expects the platform to carry high-volume manufacturing from 2027 [6]. On 8 September 2026 it announced with TSMC a move to 12-inch photomasks to lift the field-size limit, with a pilot mask line in 2031 and production systems by 2033 [10].

EUV systems ASML booked as revenue each year TOOLS



Source: ASML annual reports 2024 and 2025

Who makes it

ASML took EUR 32.7 billion in net sales in 2025 at a 52.8% gross margin [1], and in July 2026 raised its 2026 forecast to EUR 43 to 45 billion at 54 to 56% [11]. It assembles more than it makes, listing 5,100 suppliers [12]. Four carry the EUV machine:

- **Zeiss SMT**, Oberkochen, makes the illumination system and the six-mirror projection optics [13].
- **Trumpf**, Ditzingen, makes the carbon dioxide drive laser, which amplifies a few watts to 40 kilowatts [14].
- **ASML San Diego**, the former Cymer, designs the source where the laser turns tin into plasma, and builds the droplet generator [15].
- **VDL ETG** builds the frames that suspend and position the mirrors [16].

We think Zeiss mirror polishing and coating, not final assembly in Veldhoven, caps EUV output.

Nikon and Canon are specialists now. Nikon sold 22 new scanners in the year to March 2026, into a market it sizes at 570 units, and lost money on Precision Equipment [17]. Canon's bet is nanoimprint: the FPA-1200NZ2C stamps the pattern instead of projecting it, down to a 14 nm linewidth [18]. No leading-edge logic customer has taken it up.

ASML net system sales by technology, 2025 %



EUV **48%** ArF immersion **42%** KrF **4%** Metrology and inspection **3%** ArF dry **2%** i-line **1%**

Source: ASML Q4 2025 investor presentation

Controls and China's answer

Export controls bar EUV from Chinese foundries, and CSIS judges the technology out of reach there despite state spending [19]. Below EUV, the restrictions arrived in this order:

- **Dutch national licensing** of advanced DUV took effect on 1 September 2023, and The Hague widened it on 6 September 2024 [20].
- **A partial licence revocation** disclosed on 1 January 2024 stopped NXT:2050i and NXT:2100i shipments to a few Chinese customers [21].
- **The BIS Affiliates Rule** extended entity-list controls to 50%-owned subsidiaries from 29 September 2025 [22], then BIS suspended it to 9 November 2026 [23].

The controls show in the mix: China was 41% of ASML's net system sales in 2024 and 33% in 2025 [3].

On DUV, China's answer is real. Reuters reported in July 2026 that the Shanghai state firm Aishengna has started producing home-grown immersion DUV tools [24]. SMEE, the incumbent, is strongest in i-line and even there holds about four percent of the world market [19].

On EUV the record is thinner, and claims of Huawei mass production in 2026 remain unverified. Reuters reported in December 2025 that a Shenzhen team of former ASML engineers had built a prototype that makes extreme ultraviolet light but no chip [25]. Between that light and a printed chip sit the mirrors, and only Zeiss can polish them.

ASML net system sales by the region tools shipped to, 2025 %



China **33%** South Korea **25%** Taiwan **22%** United States **12%** Japan **5%**

Rest of Asia **2%** EMEA **1%**

Source: ASML Q4 2025 investor presentation

The chokepoint

Lithography scores a 5 because the failure mode is total: if ASML stops shipping, nothing at the leading edge gets printed. Substitution takes decades, and even ASML could not rebuild itself quickly, since Zeiss makes its optics and Trumpf its drive laser.

For AI accelerators, EUV is the point of control. It prints the finest layers, the ones that set how densely transistors and wires pack, and TSMC expects the number of layers needing High-NA to rise as AI designs grow more complex [10]. Every added layer is more time on a tool nobody else builds.

Key evaluation criteria

- **Resolution** – set by wavelength divided by numerical aperture. 13 nm at 0.33 NA, 8 nm at 0.55 NA [7].
- **Throughput** – wafers an hour at a stated dose. 220 on the NXE:3800E, 175 on the EXE:5200B [6].
- **Overlay** – how precisely one layer lands on the one below; ASML credits the EXE:5200B's gain to new Zeiss optics [6].
- **Field size** – the patch of wafer one exposure covers. High-NA halves it, so big AI dies must be stitched until 12-inch masks arrive [7].
- **Source power** – brighter light, more wafers an hour, which is why the 1,000-watt demonstration matters [6].

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CHAPTER
05
STAGE 5 OF 15

Photomasks and Pellicles

A business worth tens of billions of yen sits under every leading-edge chip. Two Japanese firms make almost all the blank plates that EUV masks are built on, one Japanese firm inspects them, and the dust cover meant to protect them is still not in production.

IN PLAIN TERMS

A photomask is the printing plate for one layer of a chip. One plate prints the same layer on wafer after wafer, so it shapes millions of chips. No glass is transparent to the short light now in use, so the plate is not a window but a mirror of about forty pairs of ultrathin layers, and any speck of dust on it repeats on every wafer it prints. Two Japanese firms make almost all of those mirrors. An obscure business almost nobody has heard of can hold up every leading-edge chip in the world.

> **IN SHORT**
>
> Photomasks are the cheapest chokepoint in this chain, which is why nobody has funded an entrant. Three blank suppliers in the world on AGC's count [6], one of them describing its own share as exceptionally high [8], and a single-vendor actinic inspection tool [10] sit under every EUV layer of every leading-edge accelerator. Pellicles remain unproven in production, so fabs trade throughput against defect risk product by product [11]. And mask-set costs of $10 million to $40 million decide which AI chips get designed [14].

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

5-10 years

PRICE OR MARKET SIZE

A leading-edge mask set runs roughly $10M to $40M, depending on the node and on whose estimate you take

WHO LEADS

- JP **AGC** Ranks itself No.1 worldwide in EUV mask blanks and counts three suppliers in all; targeted over ¥40bn of blank sales by 2025
- JP **Hoya** Says its share of mask blanks is exceptionally high, and expects to keep its lead as customers add second sources
- JP **Lasertec** The only supplier of EUV mask inspection at the wavelength the scanner itself uses; JPY 230.5bn revenue in the year to June 2026
- AT **IMS Nanofabrication** Intel calls it the established industry leader in multi-beam mask writers, the machines that cut the pattern
- US **Photronics** Largest mask maker selling to outside customers; $849.3M revenue in fiscal 2025

WHERE IT IS MADE

- JP **Japan** Mask blanks (AGC, Hoya), DUV blanks (Shin-Etsu), inspection (Lasertec), writers (NuFlare), pellicles (Mitsui Chemicals)
- AT **Austria** IMS Nanofabrication multi-beam mask writers, Vienna
- DE **Germany** Zeiss mask repair and metrology
- US **United States** Photronics merchant mask shops, KLA inspection
- TW **Taiwan** TSMC's in-house mask shop, and a 10% stake in IMS

WHY SUBSTITUTION IS SLOW

AGC and Hoya took two decades to get EUV blanks clean enough, and Lasertec's inspection tools have no alternative at all.

WHERE CHINA STANDS

SMIC runs two in-house mask shops in Shanghai, but only for the older g-line, i-line, KrF and ArF tools. There is no Chinese EUV blank, no Chinese tool that inspects a mask at EUV wavelength and no Chinese multi-beam writer.

WHERE THE US STANDS

Photronics is the largest mask maker selling to outside customers and KLA supplies inspection tools, but no US firm makes an EUV blank or an EUV-wavelength inspection tool.

Every EUV layer on an AI accelerator needs a flawless mask, built on a blank plate that two Japanese firms supply and passed by an inspection tool that one Japanese firm makes. AGC, one of the two, says it is the only company in the world that takes an EUV blank all the way from the glass to the finished coating [1].

How it works

A deep ultraviolet mask lets light through: quartz carrying a light-blocking film such as molybdenum silicide, and Shin-Etsu supplies those blanks [2]. An EUV mask reflects instead, about 40 molybdenum and silicon layer pairs, 7 nm to a pair, laid on low-expansion glass under a tantalum absorber that carries the pattern [3]. Making the plate a mirror creates its own problems:

- **Mask 3D effects.** Light strikes the mirror at an angle rather than head-on, which distorts the printed shape, so software has to correct for it in advance.
- **Write precision.** Every edge of the absorber has to sit within a few nanometres of where the design says.
- **Particles.** Anything on the reflector prints on every wafer until someone finds it.

High-NA scanners halve the patch of wafer one exposure covers and double the exposures needed, so a large die has to be stitched from two masks [4]. On 8 September 2026 ASML and TSMC announced a move to 12-inch masks to remove that limit, with a pilot line in 2031 [5].

Variants and trade-offs

Blanks

The blank is where two decades of learning sit: laying down forty layer pairs across a six-inch plate, even to within less than an angstrom, a tenth of a nanometre, with no particle buried anywhere underneath. Nobody publishes a market share for the business, but AGC counts the suppliers. It says it is one of three companies worldwide that supply EUV blanks, and the only one that handles every material and process itself [6]. Its own data book lists EUV photomask blanks among the products where AGC holds the No.1 position worldwide [7].

Hoya says its share of the mask blank market is exceptionally high, and expects customers to move gradually toward multi-sourcing EUV blanks [8]. AGC raised its EUV blank capacity by about 30 percent in 2025, on a line in Fukushima funded under a METI supply-chain programme, and was aiming for more than ¥40 billion of sales in the business [1].

Writers

An electron-beam writer cuts the pattern into a sensitive coating on the blank. Multi-beam writers fire many beams at once, which is what lets a dense EUV mask finish in a usable time, and two firms make them: IMS Nanofabrication of Vienna and Japan's NuFlare. Intel calls IMS the established leader, and sold roughly 10% of it to TSMC in September 2023 at a valuation near $4.3 billion [9].

Inspection and repair

A mask defect costs yield on every wafer that mask prints, so fabs check masks three times: as blanks, after patterning and again in production. Actinic inspection looks at 13.5 nm, the wavelength the mask itself sees, and Lasertec is the only supplier, on JPY 230.5 billion of net sales in the year to June 2026 [10]. KLA sells optical reticle inspection; Zeiss sells e-beam repair.

Pellicles

A pellicle is a thin membrane on a frame above the mask. Particles land on it rather than on the pattern, too far out of focus to print.

EUV makes that hard. ASML's membrane is 13 nanometres thick, has to hold together at 500C in vacuum, and every photon it absorbs is light that never

reaches the wafer [11]. The light crosses it twice, on the way in and on the way back off the mirror, so absorption counts double.

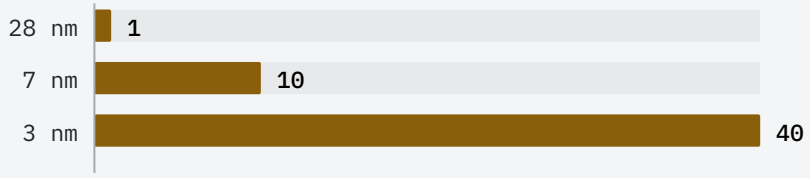
Carbon nanotube membranes are the intended fix, still short of production in a leading-edge fab. Imec has mounted them on reticles and exposed them in an NXE:3300 scanner, measuring single-pass transmission up to 97 percent and finding the effect on imaging low and correctable [12]. Mitsui Chemicals, licensed by ASML, is building capacity at Iwakuni-Ohtake for membranes above 92 percent transmittance that can stand sources above 1,000 W [13].

Cost and design economics

SemiAnalysis put mask sets beyond $1 million at 28 nm, beyond $10 million at 7 nm and near $40 million at 3 nm, as of 2022 [14].

That cost decides who can use a node at all. A GPU spread over hundreds of thousands of units absorbs a $40 million mask set easily; the same set kills a custom inference chip shipping in the thousands [14]. That is why so much AI silicon converges on a handful of designs.

Cost of one mask set by node, 2022 estimates $M



Source: SemiAnalysis

Who makes it

Mask making splits between fabs that make their own and firms that sell to everyone. TSMC, Samsung and Intel run their own leading-edge mask shops, because they want to control turnaround time. Photronics is the largest outside supplier, on $849.3 million of fiscal 2025 revenue, of which $615.1 million was IC masks [15]. Toppan, Dai Nippon Printing and Hoya take most of the rest.

SMIC runs two in-house mask shops in Shanghai, for g-line, i-line, KrF and ArF tools [16]. There is no Chinese EUV blank, no Chinese actinic inspection tool and no Chinese multi-beam writer.

The chokepoint

This stage scores a 5 for the reason lithography does, on a fraction of the money. Three firms supply EUV blanks worldwide on AGC's own count [6], one firm certifies them, and neither business is big enough to attract a funded entrant. Nobody starts from scratch at ¥40 billion of annual sales, to serve a handful of customers, with twenty years of defect learning ahead [1].

Pellicles are the unsolved part. A decade on, no membrane is transparent, tough and long-lived at once, so fabs trade throughput against defect risk. We think a production carbon nanotube pellicle would move EUV cost per wafer more than any scanner improvement.

Key evaluation criteria

- **Blank defectivity** – particles buried in the multilayer cannot be repaired, so blank yield sets mask yield.
- **Write time and placement** – multi-beam writers make dense, curved EUV patterns affordable, and Intel calls IMS the established industry leader in them [9].
- **Actinic inspectability** – only a 13.5 nm inspection sees what the scanner sees, and only Lasertec sells one [10].
- **Pellicle transmission and lifetime** – transmission counts twice, and the best demonstrated carbon nanotube membranes pass 97% in a single crossing [12].
- **Mask-set cost per node** – $10M to $40M at the leading edge, which decides which designs can use it [14].

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CHAPTER
06
STAGE 6 OF 15

Deposition and Etch

Lithography prints the pattern; deposition and etch build the chip itself, over a thousand steps per wafer. This is the one tool segment where China has taken real share.

IN PLAIN TERMS

Deposition and etch build the chip itself, like a layer cake: lay down a film a few atoms thick, print a pattern on it, burn away everything the pattern does not protect, then do it again, more than a thousand times over. The film has to reach the bottom of a hole hundreds of times deeper than it is wide, and the burn has to eat one material and leave the one beside it untouched. Four firms in America, Japan and the Netherlands make most of these machines, and China has come closer to matching them here than anywhere else.

IN SHORT

Deposition and etch are the volume business of a fab: over a thousand steps per wafer, four American, Japanese and Dutch firms taking most of the money, and no monopoly like ASML's in lithography. The AI roadmap raises difficulty faster than tool count, because gate-all-around, HBM and backside power need better selectivity and deeper holes rather than more chambers. China has closed the gap fastest here and slowest in ALD, the tool the next two logic nodes depend on.

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

High

SUBSTITUTION TIME

5-10 years

PRICE OR MARKET SIZE

No public list prices

Global semiconductor equipment billings were $135.1B in 2025, with wafer processing up 12%.

WHO LEADS

- US **Applied Materials** $28.4B revenue, fiscal 2025; leader in physical and chemical vapour deposition and in epitaxy
- US **Lam Research** $23.2B revenue, fiscal 2026; leader in dry etch and in metal atomic layer deposition
- JP **Tokyo Electron** 23% of dry etch and 38% of chemical vapour deposition, 2025
- NL **ASM International** €3.2B revenue, 2025; leader in single-wafer atomic layer deposition
- CN **Naura** 5th largest equipment vendor worldwide, 2025 estimate

WHERE IT IS MADE

- US **United States** Applied Materials (Santa Clara), Lam Research (Fremont)
- JP **Japan** Tokyo Electron, Kokusai Electric, Hitachi High-Tech
- NL **Netherlands** ASM International, Almere; ALD and epitaxy
- KR **South Korea** Large buyer; Semes and local suppliers in wet processing
- CN **China** Naura, AMEC, Piotech; largest equipment market at $49.3B in 2025

WHY SUBSTITUTION IS SLOW

Chinese firms have matched metal and insulator etch at mature nodes, but at 2 nm they still cannot cut one material while sparing the next, keep particles out, or make one chamber behave like the next. Their ALD share is under 1%.

WHERE CHINA STANDS

The strongest Chinese position in any tool segment. China-based suppliers held about 11% of dry etch and 7% of deposition in 2024, but under 1% of atomic layer deposition.

WHERE THE US STANDS

Two of the three largest suppliers are American. Washington controls dry etch, atomic layer deposition and deep-hole deposition tools by export classification number, and by a rule that reaches tools built outside the United States from US technology.

Applied Materials and Lam Research, the two largest deposition and etch suppliers, booked $28.4 billion and $23.2 billion of revenue in their latest fiscal years [1] [2].

How it works

CSET puts a single chip at more than 1,000 process steps, with the part-finished product crossing international borders 70 or more times [3].

Most of those steps happen in a vacuum chamber: a radio-frequency field strips electrons off a process gas, and the charged fragments left over do the chemistry. Change the gas, the pressure, the power and the temperature, and the same hardware either adds material or takes it away. That is why the export controls here name process specifications rather than machines.

Variants and trade-offs

Deposition families

- **Physical vapour deposition** knocks atoms off a solid target and lands them on the wafer, the workhorse for metal barriers and seed layers, $2.4 billion in 2019 [3].
- **Chemical vapour deposition** reacts gases at the wafer surface to grow a film; the plasma-enhanced version runs cooler, $3.5 billion in 2019 [3].
- **Atomic layer deposition** lays down less than one atomic layer at a time, so it coats vertical walls as evenly as horizontal floors. US regulators call it the basis for 3D scaling in 3D DRAM, 3D NAND and gate-all-around logic [4]; ASM International leads it, $1.6 billion in 2019 [3].
- **Epitaxy and electroplating** grow the strained silicon-germanium source and drain, and fill the copper wiring trenches [3].

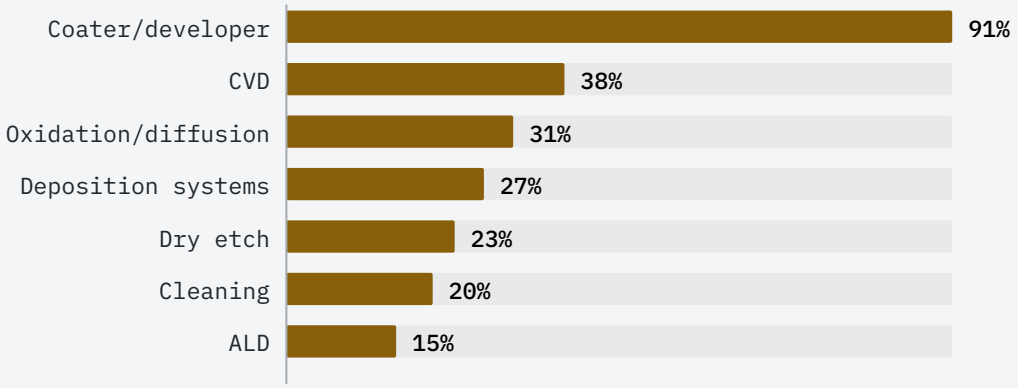
Etch families

- **Conductor etch** cuts gates, metals and silicon, $6.5 billion in 2019, led by Lam [3].
- **Dielectric etch** cuts the insulating oxides and nitrides between them, $4.3 billion in 2019, led by Tokyo Electron [3].
- **High-aspect-ratio etch** decides whether memory works at all: a 3D NAND channel or DRAM capacitor is a hole far deeper than it is wide that has to come out straight. Lam chills the wafer to sharpen the profile, and says Cryo 3.0 holds critical-dimension deviation under 0.1 percent down channels 10 microns deep [5].

AI hardware asks for sideways cuts, deeper holes and a second metal stack on the wafer's back.

- **Gate-all-around** needs isotropic etch, which cuts sideways, to release the silicon nanosheets from the silicon-germanium between them, and BIS ties its dry etch controls to exactly that [4].
- **HBM** needs holes etched straight through the silicon so chips can be stacked and wired, which BIS controls at 10:1 or deeper above 7 microns a minute [6].
- **Backside power delivery** adds thinning, backside vias and a second metal stack. Applied Materials expects it to add about $1 billion per 100,000 wafer starts a month to its wiring business [7].

Tokyo Electron world market share by tool type, 2025 %



Source: Tokyo Electron FY2026 Q4 results presentation

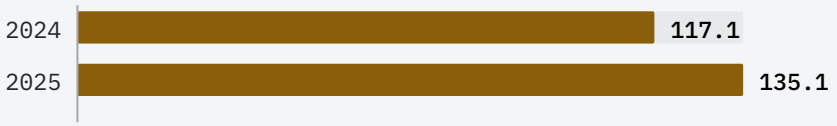
Who makes it

SEMI put global equipment billings at $135.1 billion in 2025, up 15 percent from $117.1 billion, with wafer processing equipment up 12 percent [9].

Within deposition and etch there is no single ASML. The share chart in Tokyo Electron's own results presentation puts it at 23 percent of dry etch, 27 percent of deposition systems and 15 percent of ALD in 2025, against 91 percent of coater/developer [8].

ASM International is the specialist. Its 2025 revenue reached a record EUR 3.2 billion as customers built 2 nm gate-all-around capacity, with molybdenum ALD entering volume production [10].

What the world spent on chipmaking equipment $B



Source: SEMI, Worldwide Semiconductor Equipment Market Statistics

China and the controls

China is the largest customer and the fastest-improving competitor. It bought $49.3 billion of equipment in 2025, still the biggest single market, though its spending fell half a percent while Taiwan's rose 90 percent to $31.5 billion [9]. China still takes a large share of every major vendor's sales.

- **Tokyo Electron**: 34.1 percent of sales in the year to March 2026, down from 41.7 percent [8].
- **Lam Research**: 26 percent in the June 2026 quarter, behind Taiwan at 27 percent [2].
- **Applied Materials**: 28 percent in the third fiscal quarter of 2026, down from 35 percent [11].
- **ASM International**: more than 30 percent of 2025 revenue [10].

Between 2019 and 2024 China-based suppliers went from 2 to 7 percent of deposition and from under 3 to about 9 percent of etch and clean, 11 percent in dry etch alone. In ALD they gained nothing and stayed under 1 percent [12]. AMEC, the biggest of them, turned over RMB 12.385 billion in 2025, up 36.6 percent, and spent RMB 3.744 billion on research, 30.2 percent of sales [13].

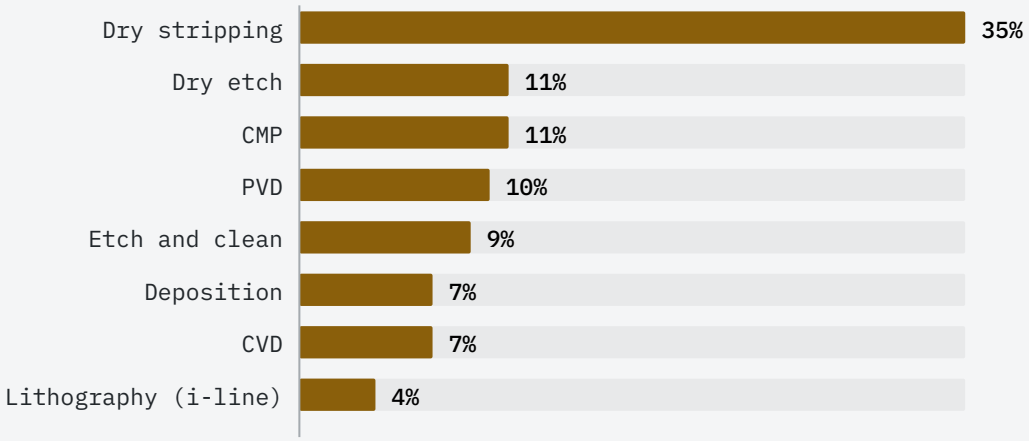
The controls tightened in stages.

- **October 2022.** The 7 October interim final rule put manufacturing equipment and US person support under licence and widened foreign-produced item rules for 28 listed Chinese entities [14].

- **October 2023.** The rule effective 17 November rewrote the equipment entries, adding isotropic and anisotropic dry etch (ECCN 3B001.c.1), spatial ALD (3B001.d.9), low-fluorine tungsten ALD and CVD (3B001.d.10) and carbon hard mask PECVD (3B001.d.5) [4].
- **December 2024.** BIS controlled 24 more types of equipment and three types of software, added 140 entities and wrote a foreign direct product rule that reaches tools built outside the United States [15]. The new entries take in TSV etch and deposition into features deeper than 200:1 [6].
- **September 2025.** The Affiliates Rule extended Entity List restrictions to any company at least 50 percent owned by a listed entity [16]; BIS then stayed it to 9 November 2026 [17].

The rules reach the vendors too: Applied Materials took a $253 million charge to settle an export controls matter [11].

China-based suppliers' share of global tool segments, 2024 %



Source: CSET, Inside Beijing's Chipmaking Offensive

The chokepoint

This is a real chokepoint but a softer one than lithography. Three or four credible vendors exist for most etch and deposition steps. The narrow points are ALD, where China holds under 1 percent [12], and the extremely deep, narrow holes that HBM and 3D memory need. That is where the December 2024 rule aims [6].

Key evaluation criteria

- **Selectivity** – how hard the process attacks one material and spares another. BIS controls wet processing at a silicon-germanium-to-silicon ratio of 100:1 [4].
- **Conformality and aspect ratio** – whether a film reaches the bottom of a hole, and how deep that hole is against its width. Controls start at 10:1 for TSV etch and 200:1 for 3D DRAM deposition [6].
- **Uniformity** – variation across a 300 mm wafer, with under 2 percent treated as an advanced-manufacturing marker [6].
- **Throughput** – wafers per hour, which sets cost per layer and decides how many lots a fab can run.
- **Particles and chamber matching** – how much of the defect budget these tools eat, and whether one chamber behaves like the next; the tools in the next chapter measure it.

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CHAPTER
07
STAGE 7 OF 15

Metrology and Inspection

Nobody builds a fab around process control, and no fab works without it. One American firm holds about 57% of the market, roughly seven times its nearest rival, and is still pulling away.

IN PLAIN TERMS

Metrology and inspection are the factory's quality control. Between the steps that build a chip, another machine looks the wafer over: are the lines the right width, did this layer land on the last one, has a speck of dirt killed a circuit. On a die as big as an AI accelerator, halving the stray specks lifts the share of working chips from about half to about seven in ten. One American firm sells seven times as many of those machines as its nearest rival, so a fab cut off from it can buy every other tool and still not learn why its chips fail.

IN SHORT

Yield is set by defect density and die area, and AI accelerators have the largest dies in production. KLA holds about 57 percent of the market, six to seven times its nearest rival, and is still gaining [2], with Applied Materials, Lasertec, Hitachi High-Tech, ASML, Nova, Onto and Camtek splitting the rest [3]. KLA's packaging process control revenue is heading toward $1.1 billion in 2026 [1]. And this is the one tool segment where nobody has recorded a Chinese gain, which is why the controls on it are the hardest for China to work around.

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

High

SUBSTITUTION TIME

10+ years

PRICE OR MARKET SIZE

A $13.4B process control tool market in 2024, about 57% of it KLA's

KLA's fiscal 2026 revenue was $13.58B against a wafer equipment market of roughly $120B in 2025, so process control is around a tenth of what a fab spends on tools.

WHO LEADS

- US **KLA** 56.5% of process control in 2024 on its own investor-day count, 6.5 times its nearest rival; $13.58B revenue, fiscal 2026
- US **Applied Materials** 10.5% of process control tools in 2024: electron-beam defect review, line-width measurement and film metrology
- JP **Lasertec** 7.7% of process control tools in 2024, almost all of it mask inspection
- NL **ASML** 5.2% of process control tools in 2024: YieldStar layer alignment and HMI multibeam electron inspection
- IL **Nova** 3.7% of process control tools in 2024; $880.6M revenue, 2025
- US **Onto Innovation** 3.0% of process control tools in 2024; $1.01B revenue, 2025

WHERE IT IS MADE

- US **United States** KLA (Milpitas), Onto Innovation (Wilmington, MA), Applied Materials
- NL **Netherlands** ASML YieldStar and HMI e-beam, Veldhoven and San Jose
- JP **Japan** Hitachi High-Tech CD-SEM; Lasertec mask inspection; Rigaku X-ray
- IL **Israel** Nova (Rehovot) and Camtek (Migdal Haemek)
- CN **China** Domestic entrants only; 0.8% of the process control tool market in 2024

WHY SUBSTITUTION IS SLOW

Sensitivity comes from optics, electron columns and decades of catalogued defects. China gained real share in etch and deposition between 2019 and 2024; in process control it was still at 0.8% in 2024.

WHERE CHINA STANDS

The weakest Chinese position of any major tool segment. In 2019 domestic firms held 3.4% of line-width measurement, 1.3% of defect inspection and none of mask or packaging inspection; in 2024 they were still at 0.8% of the whole process control tool market. CSET's 2026 survey of Chinese share gains runs through deposition, etch, polishing, lithography, packaging and test, and never reaches process control.

WHERE THE US STANDS

KLA is the largest process-control supplier in the world. It puts its own 2024 share at 56.5%, or 6.5 times its nearest rival; CSET's Supply Chain Explorer puts it at 57.2% of the same year's market. US firms take 72% of the segment, Japanese firms 15%. The December 2024 rule pulled inspection and measurement tools for patterned 300 mm wafers onto the controlled list, at a threshold of 21 nm defects.

Process control is about a tenth of what a fab spends on tools, out of a wafer equipment market of roughly $120 billion in 2025 [1]. KLA takes more than half of that tenth. At its March 2026 investor day it put its own 2024 share at 56.5%, six and a half times its nearest competitor [2]; CSET, counting the same year independently, gets 57.2% [3].

How it works

A wafer loses value two ways: whole regions ruined by mishandling, misalignment or over-etching, and single circuits killed by stray material or an airborne particle [4].

The second kind follows an equation: die yield is (1 + D0A/a) to the power minus a, where D0 is defects per square centimetre, A is die area and a is a clustering constant of 3 to 4. Yield falls faster than area rises: at a defect density of 0.8 a 1 square centimetre die yields 49 percent and a 2 by 2 centimetre die 11 percent, so the bigger die costs about twenty times as much [4].

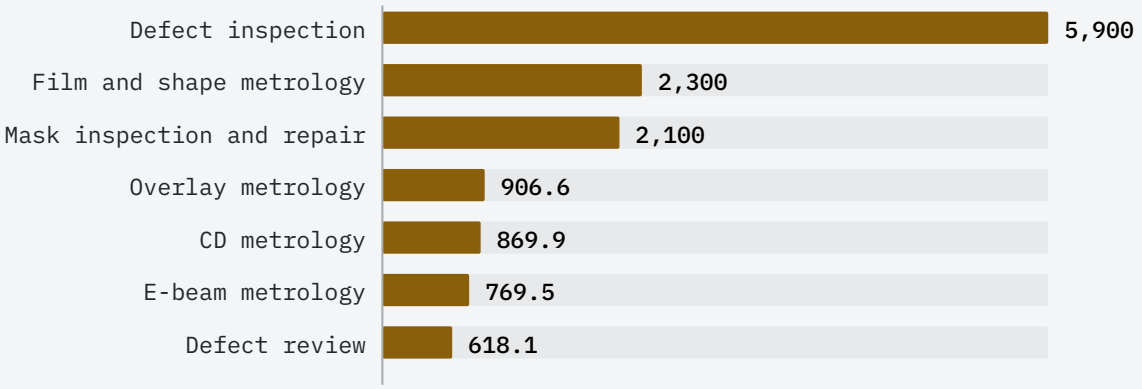
Run that equation with a clustering constant of 3 for an 8 square centimetre die, about as much as one exposure can print, and it yields about 49 percent at a defect density of 0.10 and about 69 percent at 0.05 [4]. Getting from one to the other is yield learning: find which of the thousand-odd steps is losing dies, fix it, move on.

Variants and trade-offs

The tools trade sensitivity against speed. All the market sizes and shares below are CSET's, for 2024 [3].

- **Defect inspection** scans a whole wafer with light or electrons and compares each die against its neighbours. It is the largest family by a distance, $5.9 billion, and the one KLA owns outright at 84.6 percent. ASML's HMI eScan 1100 runs 25 electron beams at once, up to 15 times the speed of a single beam, and finds patterning defects down to 7 nm [5].
- **Film and shape metrology** measures how thick each film is and how flat the wafer sits, without cutting it open: $2.3 billion, with KLA at 47 percent, Nova at 20.6 and Onto Innovation at 13.6. Onto paid about $720 million in August 2026 for 27 percent of Japan's X-ray maker Rigaku [6].
- **Mask inspection and repair** gets its own tools, because a defect on a mask repeats on every die: $2.1 billion, and the one family KLA does not lead, behind Lasertec at 45.6 percent against KLA's 40.7 and Zeiss's 11.
- **Overlay metrology** measures whether this layer sits on the one below: $907 million, split almost evenly between KLA at 54.2 percent and ASML at 45.1.
- **Critical dimension metrology** measures how wide a printed feature actually came out: $870 million, and the one family Japan leads, with Hitachi at 69 percent and Applied Materials at 30.
- **Electron-beam metrology** resolves detail too fine for light: $770 million, led by Applied Materials at 49.5 percent with ASML at 37.6 and KLA at 11.9.

Process control market by tool family, 2024 $M



Source: ETO Supply Chain Explorer, CSET

Who makes it

KLA is most of the segment: 57.2 percent of a $13.4 billion market in 2024 [3], up two points on its own count since 2021 [2]. Its fiscal 2026 revenue was $13.579 billion, after $12.156 billion in fiscal 2025 [7], and it says it gained share again in 2025 across mask, optical wafer and e-beam inspection [8].

Process control tool revenue share, 2024 %



KLA 57.2% Applied Materials 10.5% Lasertec 7.7% Hitachi High-Tech 5.6%
ASML 5.2% Nova 3.7% Onto Innovation 3% Camtek 1.5% Rigaku 1.2%
All others 4.4%

Source: ETO Supply Chain Explorer, CSET

The rest are specialists, and nobody else reaches 11 percent.

- **Applied Materials** is the largest of them at 10.5 percent, selling e-beam defect review, CD-SEM line-width tools and film metrology next to the deposition and etch tools that create the defects.
- **Lasertec** holds 7.7 percent, almost all of it mask inspection; **Hitachi High-Tech** 5.6 percent, almost all of it CD-SEM; **ASML** 5.2 percent through YieldStar overlay and HMI e-beam inspection.
- **Nova** holds 3.7 percent and took $880.6 million in 2025 on optical film and dimensional metrology [9].
- **Onto Innovation** holds 3.0 percent on $1,005.3 million of 2025 revenue [10] and **Camtek** 1.5 percent on $496.1 million [11].

Patterning, which covers overlay and mask inspection, grew 61 percent year on year in the June 2026 quarter, when Taiwan took 31 percent of revenue and China 26 percent [1].

KLA revenue by product line, June 2026 quarter %



Wafer inspection 49% Services 22% Patterning 20%
PCB and component inspection 4% Specialty semiconductor process 4% Other 1%

Source: KLA letter to shareholders, Q4 FY2026

Process control's share of fab spending is still rising, and KLA credits tighter specifications, less built-in redundancy and more logic inside memory built for high-performance computing [8].

The advantage is analytics as much as optics: inspection throws up far more candidate defects than engineers can look at, and KLA has spent five years putting AI classification through its whole line [8].

Packaging, AI and China

The AI packaging build-out is where this segment is growing fastest. KLA took the number one position in advanced wafer-level packaging process control in 2025, adding 14 points [8], and expects about $1.1 billion of packaging process control revenue in 2026, more than 70 percent growth [1]. Onto signed a volume purchase agreement worth over $240 million with an HBM maker for 2D inspection and 3D bump metrology through 2027 [10].

China has made less progress here than anywhere else in the tool chain. In 2019 Chinese firms held 3.4 percent of line-width measurement, 1.3 percent of defect inspection and none of photomask or packaging inspection [12]. Five years later they were at 0.8 percent of the whole process control tool market [3]. CSET's 2026 audit of where China gained runs through deposition, etch, polishing, lithography, packaging and test, and never reaches process control [13]. Nobody has published a Chinese gain here, which is not the same as there being none.

The December 2024 rule wrote process control into the export lists. It controls defect inspection and measurement of patterned 300 mm wafers where the tool can find defects of 21 nm or smaller using light below 400 nm, or an electron beam resolving 1.65 nm, or a cold field emission source, or two or more beams [14]. A foreign direct product rule carries the same reach to tools built outside the United States [15].

The chokepoint

Process control is a chokepoint of concentration rather than of physics. Several firms can build a good film metrology tool, but one firm sets the standard across nearly the whole flow, and its lead is widening. A sanctioned fab can buy scanners and etchers on the grey market and still fail to get production going: without sensitive inspection it cannot find out why its yield is low.

Key evaluation criteria

- **Sensitivity** – the smallest killer defect a tool can find. ASML's multibeam system is specified to 7 nm patterning defects [5].
- **Throughput** – wafers per hour, which decides how many lots you can afford to look at; multibeam buys up to 15x over single-beam [5].
- **Nuisance rate** – the share of flagged events that turn out not to be real defects, which is what AI classification attacks.
- **Sampling plan** – how many wafers and sites per lot, a cost decision as much as a physics one, and what sets the price of process control.
- **Time to result** – how fast a measurement reaches the process engineer, which sets the speed of yield learning.

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CHAPTER
08
STAGE 8 OF 15

Transistors and the Front End

Three companies can build a 2 nm-class switch. Shrinking stopped paying for itself about a decade ago, and every gain since has been bought with extra process steps.

IN PLAIN TERMS

The front end of a chip factory builds the switches. Each one is a valve on a pipe: current runs along a silicon channel, and a voltage on the gate above it opens or shuts the flow. Make the channel short enough and the valve stops sealing, so current leaks through even when the gate is off and the chip burns power doing nothing. The fix is to wrap the gate around all four sides of the channel, a sequence that has to run in exact order. Only TSMC, Intel and Samsung can do it, a list short enough for an export rule to name in full.

IN SHORT

Foundries have bought every gain since the FinFET era with process steps and new materials rather than with geometry. TSMC's own numbers show the density gain per node falling from 1.2 times at N2P to about 1.1 at A16 [4]. Three companies can run a 2 nm-class process, and only TSMC runs it at volume for outside customers. SRAM has stopped shrinking, which is why accelerators keep their memory off the logic die. China can reach 5 nm-class geometry without EUV, and CSIS still reads working EUV systems as out of reach, which is the gap the export controls defend [16].

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

10+ years

PRICE OR MARKET SIZE

No foundry publishes leading-edge wafer prices, so the public signal is TSMC's own density gain per node, which falls from 1.2x at N2P to about 1.1x at A16

WHO LEADS

- TW **TSMC** N2 in volume production since 4Q25, the first nanosheet node
- US **Intel** 18A in high-volume manufacturing; 18A-P adds 9% performance at the same power
- KR **Samsung Foundry** SF2 family; SF2Z adds backside power
- CN **SMIC** N+2 and N+3, printed in several passes on older deep-ultraviolet tools, no EUV
- JP **Rapidus** State-backed 2 nm entrant targeting mass production in 2027

WHERE IT IS MADE

- TW **Taiwan** TSMC N2 at Fab 20 Hsinchu and Fab 22 Kaohsiung
- US **United States** Intel 18A at Fab 52, Arizona; TSMC Arizona on N4 and N3
- KR **South Korea** Samsung Hwaseong and Pyeongtaek
- JP **Japan** Rapidus Chitose, Hokkaido
- CN **China** SMIC Shanghai and Beijing 300 mm lines

WHY SUBSTITUTION IS SLOW

A 2 nm-class process is over a thousand steps tuned to one specific set of tools; SMIC has spent a decade and tens of billions to reach a node TSMC retired in 2019.

WHERE CHINA STANDS

SMIC ships an N+3 node measured at 113.4 million transistors per square millimetre, printing each layer in two or four passes because it cannot buy an extreme-ultraviolet scanner. Its narrowest wires sit closer together than Intel 18A's, which is what a decade of state money buys without one.

WHERE THE US STANDS

Intel runs the only leading-edge transistor line owned by an American company. It brought gate-all-around and backside power to market together on 18A and is the first to run High-NA EUV in high-volume manufacturing.

A Blackwell GPU is 208 billion switches on two dies, each the largest a lithography machine can print in one shot [1]. Every switch has to turn on hard, turn off completely, and keep doing it for years. Making the switches is the front end of line; wiring them together is the back end.

How it works

That leak is short-channel leakage, and it has three fixes:

- **Better insulator.** Thicker, so less current tunnels through it, yet gripping the channel as tightly as before: Intel switched the gate insulator from silicon dioxide to hafnium at 45 nm in 2007 [2].
- **More gate.** Wrap gate metal around more sides of the channel, as fins and nanosheets do.
- **Thinner channel.** Thin the silicon until no current can flow where the gate cannot reach it.

Variants and trade-offs

Planar and FinFET

A planar transistor lays the gate on one side of a flat channel, and loses control as the channel shortens. A fin stands the channel on edge so the gate covers three sides, but current then comes in whole fins: a designer who needs more adds one, and nobody gets half a fin.

Gate-all-around nanosheets

A nanosheet lays the channel back down, stacks two to four thin sheets and threads gate metal around each one. The sheets can be drawn any width, so current comes in fine steps again, which TSMC sells as NanoFlex.

N2 entered volume production in the fourth quarter of 2025, TSMC's first nanosheet node [3]. TSMC rates the follow-on N2P at 18 percent more speed at the same power, 36 percent less power at the same speed, and 1.2 times the logic

density of N3E [4]. Intel's RibbonFET shipped on 18A, and 18A-P adds another 9 percent performance at the same power [5].

Those numbers are vendor claims until someone measures the silicon. Each foundry picks its own baseline, and none publishes a density figure you can set against a rival's.

TSMC's published density gain, node over node X VERSUS THE BASELINE NODE



Source: TSMC advanced technologies, HPC platform

Backside power delivery

Power and signal share one stack of copper wiring above the transistors, where wide power rails crowd out the signal wires. Backside power grinds the wafer thin and builds a second network underneath it.

Intel shipped it first as PowerVia: on 18A, an 11 percent cut in routed area and a tenfold cut in dynamic voltage droop, the sag in supply voltage when a block of logic switches on at once [5]. TSMC's Super Power Rail lands that network straight on each transistor's source and drain rather than on a buried rail, which is harder to build and buys more: against N2P, A16 gives 8 to 10 percent more speed, 15 to 20 percent less power and up to 1.10 times the chip density [4].

TSMC's logic roadmap beyond N2, from its own announcements

- 2028-01-01
  N2U, 3-4% faster or 8-10% lower power than N2P
- 2028-01-01
  A14, up to 15% faster and more than 20% denser than N2
- 2029-01-01
  A13, a direct shrink of A14 for 6% area savings
- 2029-01-01
  A12, the A14 platform with Super Power Rail backside power

Source: TSMC, 2026 North America Technology Symposium

CFET and what comes after

A logic gate needs two transistors that switch on opposite signals, and today they sit side by side. The research institute imec expects complementary FETs to stack them and roughly halve the pair's area beyond the 1 nm nodes [7].

The back end of line

Wiring is now as hard as the transistor. Copper seeps into the insulator around it, so every wire needs a barrier, a liner and a cap, and once the narrowest wires sit less than 20 nm apart those wrappers take a large share of the wire's cross-section [8]. Two metals are moving into those narrow levels.

- **Ruthenium** needs no barrier, so the whole wire conducts; imec has demonstrated ruthenium lines at a 16 nm pitch [9].
- **Molybdenum** conducts worse in bulk, 5.3 against copper's 1.68 microhm-cm, but in a thin film it beats tungsten by up to 30 percent and needs no barrier either, which is moving it into the contacts on each transistor [10].

Who makes it

TSMC, Intel and Samsung build gate-all-around nodes with extreme ultraviolet lithography. SMIC builds 7 and 5 nm-class nodes without it. State-backed Rapidus targets 2 nm mass production in 2027 on 267.6 billion yen raised in February 2026 [11].

Node names describe nothing physical. SMIC's N+3, sold as 5 nm-class, sets its narrowest wires 32.5 nm apart, tighter than Intel 18A's 36 nm, and reaches 113.4 million transistors per square millimetre against 107.7 for TSMC's N6 [12].

AI accelerators sit a node behind the leading edge: a die that large comes out working only on a process that has run for years. Blackwell uses a custom TSMC 4NP [1].

SRAM, the fast memory on the logic die, is the scaling failure that hurts AI chips most. TSMC claims 22 percent more SRAM density from N3E to N2, but the gain comes from the circuits around the array rather than from the cell that holds one bit [13], which is much of why accelerators push their memory off the die. See Memory and HBM.

The chokepoint

No single machine makes the front end a chokepoint. The scarce thing is the recipe: which steps, in which order, on which tools.

China shows what that recipe costs. The money is there: the third National Integrated Circuit Fund raised 344 billion yuan, about $47 billion, in May 2024 [14]. The tools are not. A Shanghai state firm began building domestic immersion scanners in 2026, in single-digit annual volumes [15]. Those are deep-ultraviolet machines, a generation behind EUV, and SMIC gets fine features out of them by printing each layer in several passes. A handful of new scanners a year keeps those lines running and comes nowhere near replacing ASML. EUV is further off again: China has announced $43 billion for domestic development, and the Center for Strategic and International Studies still reads working systems as out of reach [16].

See Lithography and Geopolitics.

Key evaluation criteria

- **Electrostatic control** – how far the gate wraps the channel. It sets how much the switch leaks when off, and the lowest supply voltage it can run on.
- **Density** – transistors per square millimetre, comparable only inside one foundry's own numbering and for one kind of cell.
- **Interconnect resistance** – below a 20 nm wire spacing the wire rather than the transistor sets the delay [8].
- **Power delivery** – the voltage lost carrying power across the die. Backside power buys back area and cuts the droop tenfold [5].
- **SRAM scaling** – the cell that holds one bit has stopped shrinking, which caps how much cache an accelerator can afford.
- **Cost per transistor** – no foundry publishes wafer prices, so the shrinking density gain per node is the proxy. We think this is the first era where cost per transistor rises.

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CHAPTER
09
STAGE 9 OF 15

Foundries and Fabs

TSMC runs six plants that each print more than 100,000 wafers a month. All six are in Taiwan, and they hold 13 of the 17 million wafers a year the company can make.

IN PLAIN TERMS

A foundry is a print shop for chips: other companies send in designs, and the foundry runs them through the same building on the same tools. The building costs tens of billions of dollars, the air inside is cleaner than an operating theatre, and a wafer crosses hundreds of machines over several months without ever touching that air. No design house sells enough of one product to fill a shop like that, so the world shares a handful of them. TSMC runs the biggest, and all six of its largest plants sit in Taiwan, which is why one island's politics reaches the whole chip supply.

IN SHORT

The foundry model won because leading-edge fabs are too expensive for one design house, and it concentrated because only the leader earns enough to build the next node. TSMC's six gigafabs hold 13 of the 17 million wafers a year it can make, and all six are in Taiwan [1]. Intel and Samsung are credible technically and unproven commercially, and Intel is now part-owned by its government [10]. Diversification costs $165 billion and a decade in the American case alone [9], which makes the concentration in Taiwan a fact about the 2020s and 2030s rather than a problem being solved.

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

5-10 years

PRICE OR MARKET SIZE

BCG puts the ten-year cost of owning a fab completed in 2026 at $35-43 billion

TSMC intends to spend $165 billion in the United States alone

WHO LEADS

- TW **TSMC** 77% of wafer revenue from 7 nm and below, 2Q26; six gigafabs, all in Taiwan
- US **Intel Foundry** $5.8bn revenue and a $2.089bn operating loss, 2Q26
- KR **Samsung Foundry** Hwaseong and Pyeongtaek; a minimum $17bn build-out at Taylor, Texas
- CN **SMIC** $9.327bn revenue in 2025 at 93.5% utilisation
- JP **Rapidus** State-backed 2 nm entrant, Chitose

WHERE IT IS MADE

- TW **Taiwan** TSMC's six gigafabs and UMC; N2 and N3 for every leading AI accelerator
- KR **South Korea** Samsung Foundry, Hwaseong and Pyeongtaek
- CN **China** SMIC and Hua Hong; the largest installed capacity of any region, almost all of it on older nodes
- US **United States** Intel Foundry, GlobalFoundries, TSMC Arizona, Samsung Taylor
- JP **Japan** TSMC Kumamoto, Rapidus Chitose

WHY SUBSTITUTION IS SLOW

TSMC's six gigafabs are all in Taiwan and took decades to build; Arizona, Kumamoto and Nanjing are subsidiaries, and none of them is a gigafab.

WHERE CHINA STANDS

SMIC and Hua Hong are the largest domestic foundries. SMIC turned over $9.327 billion in 2025 at 93.5% utilisation, but it cannot buy an EUV scanner, which caps it below the leading

edge. China is forecast to hold more installed capacity than any other region, and almost none of it is advanced.

WHERE THE US STANDS

Intel Foundry is the only US-owned leading-edge foundry and lost $2.089 billion on $5.8 billion of revenue in 2Q26. The federal government holds a 9.9% passive equity stake in Intel.

TSMC calls a plant a GIGAFAB once it runs more than 100,000 wafers a month, each a 300 mm disc, twelve inches across. It operates six of them, and every one is in Taiwan [1].

How it works

TSMC, founded in 1987, invented the pure-play model: it builds chips for other companies and promises never to compete with them. BCG puts the ten-year cost of owning a fab completed in 2026 at $35 to $43 billion, a third to two thirds above one finished three years earlier [2].

Every plant TSMC and its subsidiaries run came to more than 17 million 12-inch equivalent wafers a year in 2025 [3]. The six Taiwanese gigafabs hold 13 million of that [1]; Arizona, Kumamoto, Nanjing and Washington hold the rest.

Variants and trade-offs

Pure-play foundries

TSMC and UMC build only for others. Chips at 7 nm and below, meaning the newest processes, were 77 percent of TSMC's wafer revenue in the second quarter of 2026, split 33 percent at 5 nm, 30 at 3 nm and 3 at 2 nm, on revenue of $40.20 billion at a 67.7 percent gross margin [4]. No other foundry earns that margin, so none can fund the next node out of cash flow.

TSMC wafer revenue by node, 2Q26 % OF WAFER REVENUE



2 nm **3** 3 nm **30** 5 nm **33** 7 nm **11** 10 nm and above **23**

Source: TSMC 2Q26 results, filed with the SEC

Integrated device makers turned foundry

Intel and Samsung design and sell their own chips and rent out capacity too. That is harder: customers dislike handing designs to a competitor, and the house products take the best capacity. Intel Foundry booked $5.8 billion of revenue in the second quarter of 2026, up 31 percent, against a $2.089 billion operating loss [5].

National champions

SMIC, Hua Hong and Rapidus exist because a government decided they should. SMIC's revenue rose 16.2 percent to $9.327 billion in 2025 at 93.5 percent utilisation, on monthly capacity above a million wafers counted in the smaller 8-inch size, and a 21 percent gross margin [6]. Rapidus raised 267.6 billion yen in February 2026 to carry it to 2 nm mass production in 2027 [7].

Mature-node capacity

Most wafers are not leading-edge. The industry association SEMI expects foundries alone to reach 12.7 million wafers a month by 2026, with the fastest growth in China [8]. Those lines make power devices, analogue chips and microcontrollers, and compete on price rather than density.

Capacity and cost

TSMC intends to spend $165 billion in the United States, adding $100 billion in March 2025 to the $65 billion already committed in Phoenix, for three more fabs, two packaging plants and an R&D centre [9]. That is the largest foreign direct investment in US history, and it still does not move a gigafab.

Allocation matters as much as capacity. TSMC does not auction capacity; it books it years ahead against prepayments, so the queue for a node is set before the node exists. Through 2024 and 2025 the binding constraint was packaging rather than wafers. See Advanced packaging.

Who makes it

Intel is now partly state-owned. In August 2025 Washington bought 433.3 million newly issued Intel shares at $20.47, or $8.9 billion, for a 9.9 percent stake, paid out of $5.7 billion of unpaid CHIPS Act grants and $3.2 billion from the Secure Enclave programme. The stake carries no control, and it came with a

five-year warrant for a further 5 percent that Washington can exercise only if Intel drops below 51 percent ownership of the foundry business [10]. That makes the warrant an option on Intel keeping its fabs.

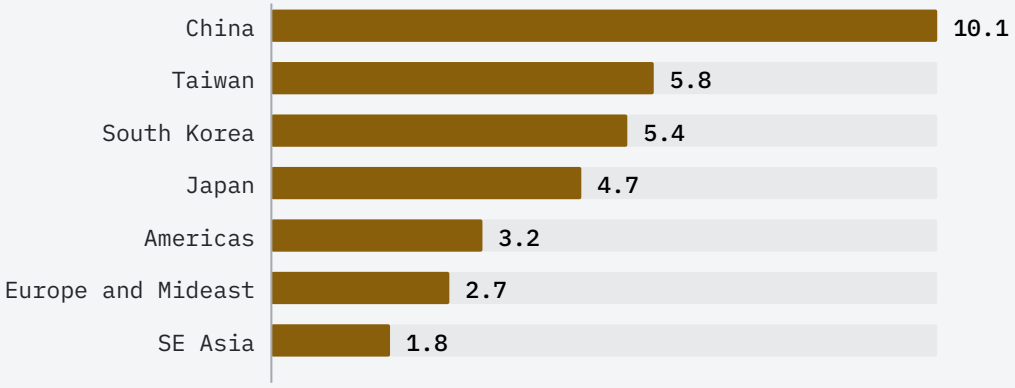
Intel's technical position improved in 2026. It reached high-volume manufacturing on part of Panther Lake using ASML's High-NA extreme ultraviolet scanners, the first company to do so, and put 18A-P into risk production, the trial runs a process goes through before it is signed off [5]. The commercial position has not caught up: 14A depends on winning external customers first.

Samsung's bet outside Korea is Taylor, Texas: a committed minimum of $17 billion, $6 billion of buildings and $11 billion of equipment, for leading-edge and mature logic [11].

The chokepoint

SEMI forecast China at 10.1 million wafers a month in 2025, the most of any region, against Taiwan's 5.8 million out of a world total of 33.7 million [8]. Almost none of China's is advanced. The United States had almost no capacity below 10 nm in 2022 and is projected to hold 28 percent of it by 2032, on $2.3 trillion of new fab investment [12].

Installed wafer fab capacity by region, 2025 forecast, all nodes MILLION 8-INCH-EQUIVALENT WAFERS PER MONTH



Source: SEMI World Fab Forecast

Three risks stack on one island, and we think only the third is unmanageable:

- **Seismic.** Taiwan sits on an active plate boundary, and a tool knocked out of calibration scraps the work in progress across a whole fab.

- **Power.** A gigafab is a large industrial load on a small island grid. We think spare generating capacity, rather than anything about silicon, sets how fast Taiwan can add fabs.
- **Political.** A blockade or conflict would remove most sub-5 nm capacity from the world market at once, with no substitute available inside a decade. See Geopolitics.

Diversification is real and slow. The hard part of a fab is a workforce that has run one process for years, and TSMC has yet to build a gigafab anywhere but Taiwan [1].

Key evaluation criteria

- **Leading-edge share** – TSMC's 77 percent of wafer revenue from 7 nm and below is why it can self-fund the next node [4].
- **Yield at ramp** – how many good chips a new process gives in its first year, which separates a profitable node from a subsidised one. Intel Foundry's loss comes from yield rather than from weak demand.
- **Ten-year cost of ownership** – $35 to $43 billion for a fab finished in 2026, and rising [2].
- **Utilisation** – how full the lines run. SMIC's 93.5 percent shows what a captive home market does for volume, and not for margins [6].
- **Customer concentration** – a foundry owned by a chipmaker competes with its own customers, which caps who will trust it with a flagship design.
- **Geographic exposure** – how much of a buyer's supply sits on one island, one grid, one fault system.

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CHAPTER
10
STAGE 10 OF 15

Memory and HBM

Three firms make every HBM stack in the world. The cube is the largest block of silicon in an AI package, and the cooling plate above it now caps its height.

IN PLAIN TERMS

High bandwidth memory is where an AI chip keeps the numbers it is working on, and it is built to hand them over fast. It stacks up to sixteen memory chips into a cube, drills copper wires straight down through them, and sets the cube beside the processor. Stacking is the hard part: each chip is ground down until the copper comes through the back, the cube must stay under a fixed height or it fouls the cooling plate, and one bad layer ruins the rest. Only SK hynix, Samsung and Micron have made it work, and a single line in an American export rule reaches all three.

IN SHORT

HBM is the most expensive component in an AI accelerator, and three firms supply all of it: the two Korean ones sold 82.5 percent in 2025 [11]. The limit now is heat and height inside a 775-micrometre cube [9], and the fix, hybrid bonding, is two generations away. The export control on HBM works because one bandwidth-density threshold catches every part worth buying [15].

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

5-10 years

PRICE OR MARKET SIZE

An HBM cube's data interface is 16 times wider than a DDR5 module's, and that width costs die area and stacking yield

WHO LEADS

- KR **SK hynix** 63.2% of HBM revenue, 2025; first to complete HBM4, at over 10 Gbps per pin
- KR **Samsung** 19.3% of HBM revenue, 2025; Hwaseong and Pyeongtaek
- US **Micron** 17.4% of HBM revenue, 2025; HBM4 above 11 Gbps per pin

WHERE IT IS MADE

- KR **South Korea** SK hynix Cheongju and Icheon; Samsung Pyeongtaek and Hwaseong
- US **United States** Micron headquarters and R&D; SK hynix packaging plant under construction in Indiana
- TW **Taiwan** Micron DRAM and HBM packaging in Taichung; TSMC builds HBM4 logic base dies
- JP **Japan** Micron Hiroshima, 1-gamma DRAM
- SG **Singapore** Micron HBM advanced packaging

WHY SUBSTITUTION IS SLOW

A new entrant needs leading-edge DRAM, a line that drills and thins wafers, a stacking process that keeps dies ground to tens of microns from warping, and Nvidia's approval.

WHERE CHINA STANDS

No Chinese firm ships HBM3E or HBM4 in volume. CXMT is the only credible domestic entrant, and the export controls are written to keep it that way.

WHERE THE US STANDS

Micron is the only US-headquartered HBM maker. BIS controls HBM above 2 GB/s per mm2 of package area under ECCN 3A090.c, which captures every stack now in production.

An AI accelerator is a machine for dragging numbers past arithmetic units, and arithmetic got cheap faster than the dragging did. High bandwidth memory now decides how fast a GPU runs, and three companies make all of it.

How it works

One cube presents a 1,024-bit interface across 32 independent channels, sixteen times wider than a standard DDR5 memory module [1]. Ordinary memory cannot be made that wide: the pins would not fit on a circuit board.

Answering a prompt is where the width earns its cost. Training reuses each stored number across a huge multiplication, so a chip can fetch little and compute a lot. Generating one word of an answer reads the whole model and the whole running record of the conversation, then waits on memory, and that record grows with the length of the conversation and the number of users served at once. Memory size and speed, rather than arithmetic, usually set how many people one chip can serve.

Variants and trade-offs

GENERATION	INTERFACE	BANDWIDTH PER STACK	CAPACITY PER STACK
HBM3E	1,024-bit, 16 channels (32 pseudo-channels)	over 1.2 TB/s	24 GB (8 chips), 36 GB (12 chips)
HBM4	2,048-bit, 32 channels	over 2.0 TB/s, up to 3.3 TB/s	up to 64 GB (16 chips, 32 Gb dies)

HBM3E

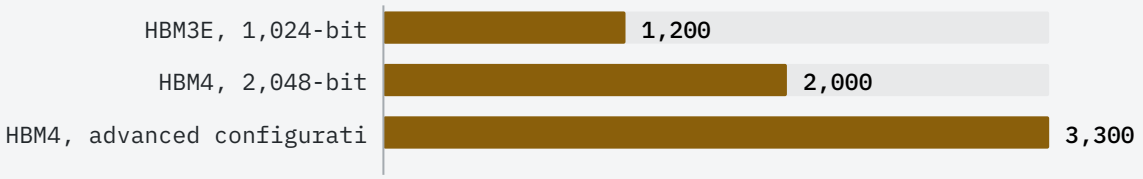
Micron ships the workhorse parts, 24 GB with eight chips stacked and 36 GB with twelve [2], and HBM3E should be about two thirds of all HBM shipments in 2026 [3].

HBM4

The standards body JEDEC published JESD270-4 in April 2025, setting the 2,048-bit interface and 2 TB/s per stack at 8 Gb/s per pin [4]. Nobody ships at that baseline: SK hynix completed HBM4 at over 10 Gb/s per pin in September 2025 [5], and Micron's parts run above 11 Gb/s for more than 2.8 TB/s per stack [6].

HBM4's other change is commercial. The chip at the bottom of the stack is now a logic chip rather than a memory one, so it has to be made in a foundry, and SK hynix partnered with TSMC to build it [7]. That hands TSMC a position inside a product it does not make.

Peak bandwidth per HBM stack GB/S



Source: Siemens EDA, HBM3E and HBM4 IC design guide, April 2026

How a stack is built

- **Drill and thin.** Etch vertical holes, the through-silicon vias, into a memory wafer, fill them with copper, then grind the wafer from behind until the copper comes through.
- **Stack and join.** SK hynix uses Advanced MR-MUF, which melts the solder bumps between stacked chips and moulds a filler around them in one pass, holding heat and warping in check [5].
- **Stay under the ceiling.** HBM4 raised the cap on a finished cube from 720 micrometres to 775, still under a millimetre, because a stack taller than the processor beside it would foul the cooling plate [9].

Hybrid bonding is the next step and it is not ready. Pressing copper pad straight onto copper pad, with no solder and no filler, would cut the spacing between

connections from about 20 micrometres to under 1, but SK hynix expects full-scale adoption only at HBM4E or HBM5, where stacks pass twenty layers [9].

Who makes it

Ordinary computer memory, DRAM, is a three-firm oligopoly, and HBM is the same three firms reordered: SK hynix, Samsung and Micron. Nobody else has both leading-edge DRAM and a stacking line inside one company.

SK hynix built its lead by being first to pass Nvidia's qualification tests, and expects roughly 70 percent of Rubin HBM4 in 2026 [3]. It broke ground on a $4 billion packaging plant in West Lafayette, Indiana in August 2026 [10], and Micron packages in Taichung and Singapore.

The split is on the record because SK hynix registered shares in the United States. Its July 2026 prospectus prints it: of 2025 HBM revenue, SK hynix took 63.2 percent, Samsung 19.3 and Micron 17.4, which puts the two Korean suppliers at 82.5 percent of the world's HBM [11]. The filing credits the research firm IDC, and the numbers move fast. In 2024 the same three stood at 56.4, 37.8 and 5.8, so Samsung halved and Micron tripled inside a year on qualification wins rather than on new capacity. By the first quarter of 2026 Micron was at 23.1 percent and Korea down to 76.9 [11].

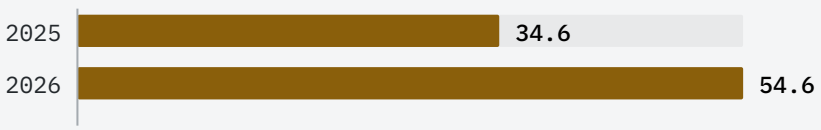
Global HBM revenue by supplier, 2025 %



SK hynix (KR) **63.2%** Samsung (KR) **19.3%** Micron (US) **17.4%**

Source: SK hynix Form 424B4 prospectus, 10 July 2026, on IDC data

HBM market revenue, Bank of America estimate $B



Source: Bank of America estimate cited in SK hynix's 2026 market outlook

Bank of America puts the 2026 HBM market at $54.6 billion, up 58 percent on the year [3]. HBM is sold forward, so that revenue is contracted long before the wafers exist, and every wafer turned over to HBM is a wafer not making ordinary DRAM. That is how AI demand raises the price of a laptop.

The chokepoint

Every leading accelerator depends on Korean or American HBM. Nvidia's GB300 NVL72 carries 20 TB of HBM3E at up to 576 TB/s [12], Rubin's VR200 up to 288 GB of HBM4 per GPU at up to 22 TB/s [13], and AMD's MI355X 288 GB of HBM3E at 8 TB/s [14].

The Bureau of Industry and Security rule of 5 December 2024 added an export classification, ECCN 3A090.c, covering HBM with a bandwidth density above 2 GB/s per square millimetre of package area, and reached stacks made outside the United States through a foreign direct product rule, which catches goods made abroad with American technology [15]. Against the JEDEC numbers, that threshold captures every stack now in production, so there is no compliant high-end HBM to sell.

China's answer is CXMT, and it is behind. We think the gap is closing from a base that did not exist five years ago, and that what stops a new entrant is the stacking line and Nvidia's approval, rather than the memory chip itself.

Key evaluation criteria

- **Bandwidth per stack** – the number that sets how fast a chip can serve answers. Over 1.2 TB/s on HBM3E, over 2 TB/s on HBM4 [8].
- **Capacity per stack** – how large a model fits on one accelerator. Up to 64 GB with sixteen chips stacked [4].
- **Stack height** – 775 micrometres for HBM4, which is what caps the number of layers before the cooling plate [9].
- **Stack yield** – one bad die scraps the cube, so yield sets the price more than wafer cost does.
- **Thermal path** – the chip at the bottom runs hottest, and every layer added above it makes the heat harder to get out.
- **Qualification** – passing Nvidia's data-rate bar, above 10 Gb/s rather than JEDEC's 8 [5].

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CHAPTER
11
STAGE 11 OF 15

Advanced Packaging

Lithography stopped being the limit on chip size in 2022. What replaced it is how large an interposer TSMC can build, and through 2025 how many it could build a month.

IN PLAIN TERMS

Packaging wires finished chips into one component and mounts them on a base built to the tolerances of a chip. A lithography machine can print only a rectangle about the size of a postage stamp in one shot, so a modern accelerator is cut into pieces and stitched back together on a shared slab of silicon, with its memory a few millimetres away. Slab, chips and base all expand at different rates when heated, so the package warps and the joints crack. Getting that right in volume is hard enough that nearly every AI accelerator is packaged by TSMC, in Taiwan.

IN SHORT

Advanced packaging capped AI chip output through 2024 and 2025: the physics was solved and the lines were not built. On Epoch AI's estimate Nvidia alone took 60.3 percent of the world's CoWoS in 2025, and four American designers took around 90 percent [9]. By March 2026 the binding constraint had moved back to front-end wafers [10]. The concentration is unchanged: China can package chips but not the 2.5D structures AI accelerators need, and the American alternatives do not start until 2028.

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

3-5 years

PRICE OR MARKET SIZE

No public wafer counts

TSMC guided investors to 60% compound annual growth in CoWoS capacity and adds more every year, but will not publish the absolute number. Epoch AI estimates Nvidia alone consumed 60.3% of the world's CoWoS capacity in 2025

WHO LEADS

- TW **TSMC** Effectively all CoWoS for leading AI accelerators
- TW **ASE Technology** NT$645.4B revenue, 2025; the largest outsourced assembly and test firm
- US **Amkor** $6.71B net sales, 2025; the largest American one
- CN **JCET** RMB 35.96B revenue, 2024; the largest Chinese one
- SG **ASMPT** $532.1M advanced packaging revenue, 2025; thermocompression bonders

WHERE IT IS MADE

- TW **Taiwan** TSMC AP fabs and the OSAT cluster; almost all 2.5D capacity for AI accelerators
- US **United States** Amkor Peoria, Arizona from 2028; two TSMC advanced packaging plants planned in Arizona
- CN **China** JCET, Tongfu and HT-Tech, strong in conventional assembly and test, weak in 2.5D
- NL **Netherlands** Besi hybrid and die bonders
- JP **Japan** Disco grinders and dicers; TEL and Shibaura bonding tools

WHY SUBSTITUTION IS SLOW

The individual steps are not exotic. What is scarce is getting a 5.5-reticle interposer and its HBM stacks to come out working, and owning enough tools to do it at the volume Nvidia alone orders.

WHERE CHINA STANDS

China's back end is far stronger than its front end. JCET alone billed RMB 35.96 billion in 2024, and Chinese suppliers gained share in packaging and test tools while gaining none in atomic layer deposition or lithography. Almost none of it is silicon-interposer 2.5D for AI accelerators.

WHERE THE US STANDS

No domestic CoWoS at volume. Amkor is building a $7B packaging and test plant in Peoria with up to $407M of CHIPS funding, production from early 2028.

A lithography scanner prints a rectangle 26 mm by 33 mm, or 858 mm2, in one exposure [1]. Every AI accelerator worth buying is bigger than that. Packaging is how the industry got around that reticle limit, and through 2024 and 2025 it capped how many accelerators the world could build.

How it works

High bandwidth memory gives a second reason to split the package: its 1,024-bit or 2,048-bit connection is too wide for a resin base board to carry, so the memory sits beside the logic on a shared silicon carrier, the interposer.

The name CoWoS says the order: chip onto wafer, then wafer onto substrate. TSMC makes a thin silicon wafer carrying wiring and vertical copper vias but no transistors, bonds logic dies and memory stacks face down onto it with microbumps 30 to 40 microns apart, then cuts out the interposer and attaches it to the resin substrate.

The individual steps are ordinary. But one cracked joint scraps a package holding eight qualified HBM stacks and two of the most expensive logic dies ever made.

Variants and trade-offs

FAMILY	CARRIER	CEILING	USED FOR
CoWoS-S	Full silicon interposer	3.3x reticle, about 2,700 mm2	H100-class parts
CoWoS-R	Resin interposer with fine redistribution wiring	Above 3.3x reticle	Cost-sensitive, fewer HBM stacks

FAMILY	CARRIER	CEILING	USED FOR
CoWoS-L	Redistribution wiring with local silicon bridges	5.5x reticle in production, 9.5x in development	Blackwell, Rubin

2.5D: silicon, RDL and bridges

TSMC has run CoWoS in volume since 2012 [2]. CoWoS-S wires most finely, but its interposer is one unbroken piece of silicon whose yield falls as the area grows, so CoWoS-L replaces the slab with fine wiring and small silicon bridges only where the wiring is dense. Intel's EMIB buries the same bridge idea in the resin substrate, and is the one credible second source when CoWoS is tight.

Blackwell was the first high-volume design on CoWoS-L, and it went badly. The bridges carrying the 10 TB/s link between the two compute dies have to be placed to a tolerance that the package's warping fights [3].

3D: SoIC, Foveros, X-Cube

TSMC's SoIC and Intel's Foveros bond dies face to face, copper pad to copper pad with no solder, packing the connections far closer than microbumps allow. TSMC put 3 nm SoIC into volume production in 2025 [4], with an A14-on-A14 version due in 2029 carrying 1.8 times as many connections between the dies as the 2 nm generation [5]. Heat decides the layout: the bottom die overheats and the top is hard to power, so 3D designs often put cache underneath rather than compute.

The escape routes

Panel-level packaging swaps the round wafer for a rectangular panel and wastes far less area; see Test and assembly. JEDEC published SPHBM4 on 13 July 2026: HBM4 memory on a base chip carrying 512 data signals instead of 2,048, which spaces the connections widely enough to mount the stack on a standard resin substrate, taking the memory off the silicon interposer altogether [6].

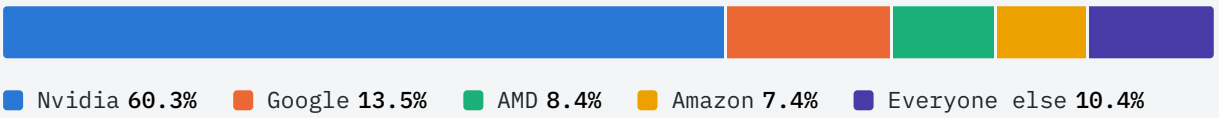
Capacity

TSMC does not publish a CoWoS wafer number, and will not. Its 2024 symposium told investors capacity would grow at a 60 percent compound rate, and on the July 2024 earnings call C.C. Wei said he could not reach supply and demand balance [7].

In October 2025 Wei would still say only that TSMC would add capacity again in 2026, give the real figure the year after, and that front end and back end were both, in his word, tight [8].

The split is public; the total is not. Epoch AI weighs each designer's component use against world supply and puts Nvidia at 60.3 percent of all CoWoS in 2025, with a 90 percent confidence interval of 56.5 to 64.3; Google took 13.5, AMD 8.4 and Amazon 7.4 [9]. Those four absorbed around 90 percent of all CoWoS and all HBM by value while taking about 12 percent of the world's advanced logic wafers, which is why packaging ran out and logic did not.

Who consumed the world's CoWoS capacity, 2025 %



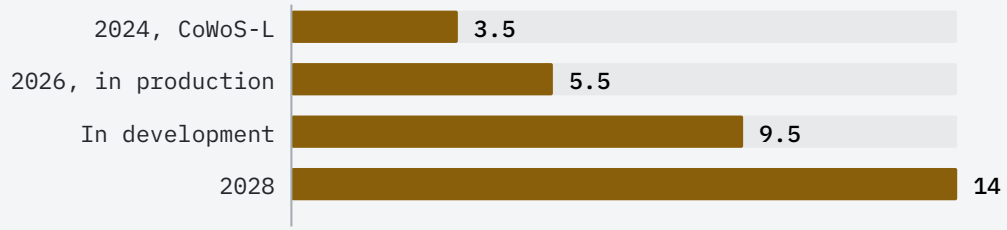
Nvidia **60.3%** Google **13.5%** AMD **8.4%** Amazon **7.4%** Everyone else **10.4%**

Source: Epoch AI, advanced packaging and HBM were the bottlenecks on AI chip production in 2025

The ordering has since flipped. SemiAnalysis reported in March 2026 that front-end wafers are now the dominant bottleneck and CoWoS is easing, partly because TSMC has no reason to build interposer lines its N3 output cannot feed [10]. The queue has eased; the concentration has not.

Who makes it

CoWoS interposer size on TSMC's roadmap RETICLES



Source: TSMC 2025 annual report and 2026 technology symposium

The word packaging covers two industries. TSMC does the 2.5D work, chips side by side on a shared carrier, in its own back-end fabs; everything else goes to outsourced assembly and test firms. ASE Technology Holding, the largest, billed NT$645.4 billion in 2025, with its assembly and test business up 23 percent, and Amkor, the largest American one, billed $6.71 billion [11] [12].

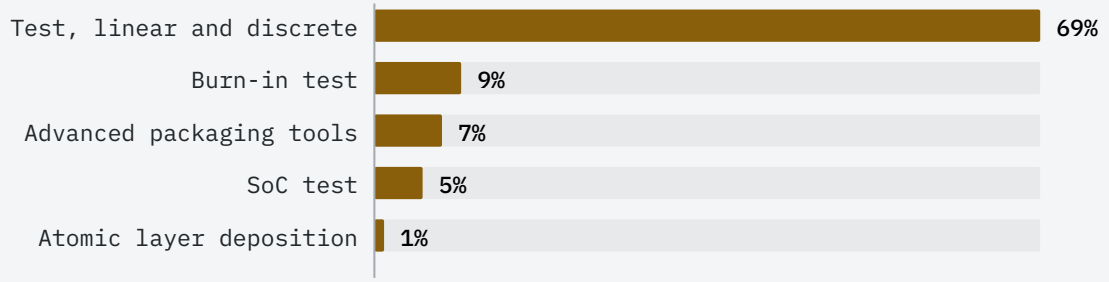
The tools are a narrower chokepoint than the assembly houses.

- **Thermocompression bonding.** Pressing dies together under heat and force. ASMPT's advanced packaging revenue reached $532.1 million in 2025, with the bonder line up about 146 percent; it targets 35 to 40 percent of a market it puts at $1.6 billion by 2028 [13].
- **Hybrid bonding.** Solder disappears and copper pads meet directly. Applied Materials built its Kinex bonder with Besi, which makes the placement head [14].
- **Grinding and dicing.** Disco dominates the steps that thin wafers and cut packages apart; see Test and assembly [15].

America is building, slowly. TSMC's Arizona commitment reached $265 billion in July 2026 and covers two advanced packaging plants alongside 10 fabs [16]. Amkor broke ground in Peoria in October 2025 on a $7 billion campus, backed by up to $407 million of CHIPS money, producing from early 2028 [17], and Wei confirms it will open before TSMC's own two do [8].

The chokepoint

China-based suppliers' share of back-end tool segments, 2024 %



Source: CSET, Inside Beijing’s Chipmaking Offensive (ALD shown for contrast, under 1%)

JCET, the largest Chinese assembler, billed RMB 35.96 billion in 2024, up 21 percent [19], and Chinese toolmakers took real share in packaging and test equipment over the same years they took none in atomic layer deposition [18]. Packaging tools are less export-controlled than lithography, and building big chips from small chiplets cut on SMIC silicon and assembled at home is the most credible route around the restrictions Huawei faces. China still has no silicon-interposer 2.5D at volume with qualified HBM beside it.

Each generation on TSMC's roadmap asks more of the interposer.

- **2028.** A 14-reticle interposer carrying about ten large compute dies and 20 HBM stacks, roughly 12,000 mm2 of silicon, about the area of a compact disc [5].
- **2029.** More than 14 reticles, and a 40-reticle System-on-Wafer beside it [5].

Key evaluation criteria

- **Interposer area** – the ceiling on how much logic and memory fit, now 5.5x reticle on CoWoS-L [5].
- **Package yield** – nobody publishes it, and it decides whether a generation ships on time.
- **Warpage control** – the expansion mismatch between die, bridge, interposer and substrate that broke early CoWoS-L and gets harder as packages grow [3].
- **HBM stacks supported** – about 20 on the 14-reticle package due in 2028 [5].
- **Bonder throughput** – thermocompression and hybrid bonding place one die at a time, slowly, and three or four firms hold the tools [13].
- **Substrate size** – a 14-reticle interposer needs a resin substrate that will not warp under it, which pushes ABF build-up film to its limits; see Substrates and PCBs.

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CHAPTER
12
STAGE 12 OF 15

Substrates and PCBs

A Japanese food company makes 95 percent of the insulating film inside high-end chip packages, and a Japanese textile company weaves the glass cloth beneath it.

IN PLAIN TERMS

A package substrate is the adapter between a chip and the circuit board under it. The chip's connections come out finer than a human hair; the board works in millimetres. The substrate is built like plywood: a stiff core of glass cloth and resin, thin sheets of epoxy film pressed onto both faces, holes burned through each sheet by laser, and copper plated into the holes. Every added layer is another hole that can fail. A handful of firms build these substrates, and the film and the glass cloth each come from a single Japanese supplier, so a shortage here stops accelerators shipping.

IN SHORT

TSMC says packaging capacity already caps what its customers can ship, and the two materials that make a substrate possible come from single Japanese suppliers: Ajinomoto puts its film at 95 percent of that market, Nitto Boseki claims an overwhelming advantage in the glass cloth beneath it. Substrate plants in Japan, Taiwan and Korea are spending billions, but film and glass cloth add capacity on a slower clock than GPUs do. Glass cores fix the physics eventually; on Intel's and Samsung's own dates, they do not fix 2027.

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

5-7 years

PRICE OR MARKET SIZE

Ibiden alone is spending about JPY 500 billion on high-performance substrate capacity from fiscal 2026 to 2028, JPY 220 billion of it in the first phase

WHO LEADS

- JP **Ajinomoto** 95% of the world market for insulating film in high-end processor packages, on its own account
- JP **Nitto Boseki** Claims an overwhelming lead in low-expansion glass cloth
- JP **Ibiden** Says it takes close to 100% of each new generation of interposer substrates at launch
- TW **Unimicron** Build-up substrates were 52% of sales in Q2 2026
- KR **Samsung Electro-Mechanics** Package solution sales KRW 771.6bn in Q2 2026, up 37%

WHERE IT IS MADE

- JP **Japan** Build-up film, low-expansion glass cloth, and Ibiden and Shinko substrate plants in Gifu and Nagano
- TW **Taiwan** Unimicron, Nan Ya PCB, Kinsus; the largest cluster of substrate makers selling to all comers
- KR **South Korea** Samsung Electro-Mechanics, LG Innotek, and a glass-core joint venture with Sumitomo Chemical
- CN **China** Most of the circuit board competition, alongside Taiwan; now pushing into glass cloth and build-up film
- AT **Austria** AT&S, the only European advanced substrate maker of scale

WHY SUBSTITUTION IS SLOW

Every substrate maker and every packaging house would have to requalify a new film or core laminate layer by layer, and Ajinomoto's film has been the reference since 1999.

WHERE CHINA STANDS

China and Taiwan hold most of the circuit board competition, and Nitto Boseki itself calls Chinese and Taiwanese firms aggressive fast followers in special glass. China still has almost no position in high-end build-up film.

WHERE THE US STANDS

No American maker of build-up substrates at scale. TTM Technologies is the main US-listed circuit board maker; data centre computing reached 24% of its sales in 2025, up from 14% in 2023. Intel is pushing glass substrates for the second half of the decade.

On TSMC's July 2026 earnings call, chief executive C.C. Wei said packaging capacity is now so tight that it limits customers' growth [1]. The silicon interposer that wires chips together inside a package spans a few millimetres. Past its edge the chain turns to epoxy and glass: a build-up substrate, a printed circuit board of thirty layers, then a rack.

How it works

The substrate maker drills the core and plates it with copper before the build-up starts. Each cycle then presses on a film of epoxy about 10 micrometres thick, burns holes through it with a laser, plates copper into the holes to link the layers, and etches the wiring on top [2]. A human hair is ten times thicker than one of those films [3].

Silicon and epoxy swell at different rates when heated, so the larger the package, the more it warps. AI parts are the largest: Ajinomoto counts 18 layers of its film in a high-performance computing substrate against six in a PC part, on a body three and a half times the area [3]. The fix is a core that barely moves, which means low-expansion glass cloth and tighter control of the resin.

Variants and trade-offs

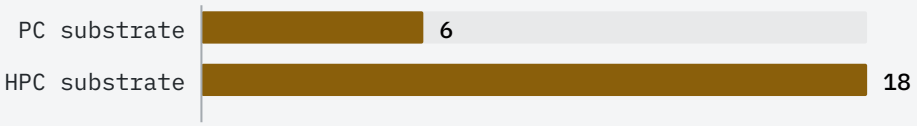
BT substrates

Bismaleimide-triazine laminate, or BT, is the cheaper and older family. It still wins on unit volume and suits memory stacks and mobile parts, but it cannot hold the fine wiring or the flatness a large logic die needs.

ABF build-up substrates

Ajinomoto Build-up Film, or ABF, is the insulating sheet between the copper layers, and it sits under every high-end CPU, GPU and AI accelerator. A major chipmaker first used it in 1999, and Ajinomoto calls it the de facto standard in package development ever since [3]. Asked on the company's own site whether its share was nearer 30 or 50 percent, the electronic materials division answered: "More like 95% market share," and said practically all high-performance computers and servers rely on it [2].

Layers of build-up film in one package substrate LAYERS



Source: Ajinomoto business briefing on ABF, June 2023

Glass core substrates

Swap the resin core for glass and most of the swelling problem goes away, which is why everyone is trying.

- **Intel** announced glass substrates in September 2023 for the latter part of this decade [4].
- **Samsung Electro-Mechanics** signed a joint venture with Sumitomo Chemical in November 2025 to make glass core, with mass production after 2027 [5].

Both programmes point past 2027, so glass answers the next problem rather than this one.

The materials underneath

Ajinomoto's film is a rounding error in a food company's accounts and a chokepoint in everyone else's. The Healthcare and Others segment that carries electronic materials lifted business profit 45.1% to JPY 66.2 billion in the year to March 2026, and Ajinomoto credits the materials for most of that [6]. Capacity comes slowly: about JPY 25 billion of growth investment from fiscal 2023, sized against demand out to 2030 [3].

One layer down, the core tells the same story. Nitto Boseki's T-glass is stiff and barely swells when heated, which is what a dense package substrate needs [7]. The

company claims an overwhelming advantage in quality and capacity for low-expansion glass, says demand has run two to three times ahead of its fiscal 2024 to 2027 plan, and is spending JPY 80 billion over those four years to catch up [8]. Apple and Qualcomm are among the customers worried about Japanese glass cloth supply [9].

China is attacking this layer, and Nitto Boseki says so itself: it names Chinese and Taiwanese firms as aggressive fast followers in special glass [8]. The prize is a chip materials market Nikkei sizes at $73 billion [10].

Who makes it

The plants that build substrates for other firms are an Asian oligopoly, and every large one is expanding. The White House's 2021 hundred-day supply chain review recorded build-up substrate among the materials commenters called vulnerable, and named the makers as Ibiden, Shinko, Nanya, Samsung, Unimicron, Shennan Circuits, Zhuhai Yueya and AKM. Not one is American [11].

- **Ibiden** will invest about JPY 500 billion from fiscal 2026 to fiscal 2028, starting with JPY 220 billion at its Gama plant and mass production from fiscal 2027 [12]. In May 2026 management said it typically takes close to 100 percent of a new generation of interposer substrates at launch and loses 20 to 30 points three to six months later. On silicon-bridge substrates, it said, wiring the underside is hard enough that it expects to keep its technical lead through fiscal 2030 [13].
- **Unimicron** took NT$42.9 billion of sales in the second quarter of 2026, 52% of it build-up substrates and 61% of it AI data centre work [14].
- **Samsung Electro-Mechanics** grew package solution sales 37% year on year to KRW 771.6 billion in the second quarter, on substrates for AI accelerators and server processors [15].

Shinko changed hands: Fujitsu sold its 50.02% stake to a fund run by the state-backed Japan Investment Corporation [16].

Unimicron sales by technology, second quarter 2026 %



ABF substrate **52%** PCB **27%** BT substrate **9%** HDI **9%** FPC **2%** Other **1%**

Source: Unimicron 2026 Q2 earnings conference

The boards under the GPU

TTM Technologies says it regularly builds printed circuit boards of more than 30 layers and can go past 70 [17]. Those layers carry hundreds of amps to the GPUs and route traffic between them fast enough that how little signal the laminate soaks up becomes a main design constraint.

Data centre computing was 24% of TTM's US$2.91 billion of fiscal 2025 revenue, against 14% two years earlier [17] [18]. It names its competition as mostly Chinese and Taiwanese [17]. The hedge is Southeast Asia, where Thailand has built an AI board industry [19].

The chokepoint

Money can expand an oligopoly of substrate plants, and it is being spent. The single points of failure are one layer down: one company's film and one company's glass cloth, both Japanese, both running flat out, both adding capacity on building schedules measured in years while AI demand compounds quarterly.

Flatness breaks first as packages grow. TSMC has told investors that its CoWoS packaging roadmap goes beyond 14 times the area a lithography machine can print in one shot [1]. Bigger bodies mean more warping, more layers, more holes that can break the circuit, and lower yield on a part that is already expensive before a die touches it.

Key evaluation criteria

- **Layer count** sets how much current and how many signals the substrate can carry. Each added layer is another chance to lose the part.
- **Body size** drives everything else. Once a body is much larger than the dies it carries, warping and handling decide what can be built, not the wiring.
- **Coefficient of thermal expansion**, how much a material swells when heated, must be close enough to silicon that the package survives repeated heating and cooling, which is what T-glass buys and why its supply matters.
- **Dielectric properties**, how much signal the core laminate soaks up, set the speeds the substrate and the board underneath can run.
- **Qualification time** is what protects the incumbents. Approving a second supplier for a film or a laminate takes years.

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CHAPTER
13
STAGE 13 OF 15

Test and Assembly

Assembly and test decide which dies are allowed into a package that costs more than a car. One Japanese firm now sells two thirds of the world's chip testers.

IN PLAIN TERMS

Before a chip can be sold, a machine has to interrogate it and check every answer. Thousands of needles land on pads on the wafer and drive test patterns into each chip; the tester marks the ones that answer wrong, and they never leave. This matters more each year because of arithmetic: a modern accelerator glues together a dozen expensive pieces, and one bad piece throws away all of them. So the industry tests earlier, hotter and longer than it used to, and two firms, one Japanese and one American, sell most of the machines that do it.

IN SHORT

Test used to be the cheap step at the end. With a dozen expensive dies in one package, it now insures everything upstream: the market for logic chip testers grew about 68% in 2025 [7] and burn-in has started moving back onto the wafer [3]. Advantest has turned a duopoly into a two-thirds share of the machines. The assembly floor beneath it stays fragmented, and four of the top ten assembly and test firms are Chinese [18], which is why export policy has little grip here.

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

High

SUBSTITUTION TIME

3-5 years

PRICE OR MARKET SIZE

ASE and Amkor alone billed US$27.3 billion in 2025

ASE spent US$4.1 billion on machinery and buildings in the first half of 2026

WHO LEADS

- JP **Advantest** 65% of the tester market in 2025
- US **Teradyne** Second in testers; about 80% of the market with Advantest
- TW **ASE** US$20.6bn of 2025 revenue, the largest assembly and test contractor
- US **Amkor** US$6.71bn of 2025 revenue
- JP **Disco** Leading share in dicing saws, grinders and blades

WHERE IT IS MADE

- TW **Taiwan** ASE, Powertech, KYEC and ChipMOS; the largest assembly and test cluster
- JP **Japan** Advantest testers, Disco dicing and grinding, Micronics Japan probe cards
- US **United States** Teradyne, FormFactor, Cohu, Aehr; Amkor's Arizona packaging plant
- CN **China** Four of the top ten assembly and test firms by 2024 revenue are China-headquartered
- MY **Malaysia** Penang and Kulim; Malaysia ships about 13% of the world's packaged chips

WHY SUBSTITUTION IS SLOW

Assembly and test floors can be built in a few years, but changing tester vendor means rebuilding the entire device development, evaluation and production test environment, so customers almost never switch.

WHERE CHINA STANDS

Assembly and test is where China is most competitive. Georgetown's CSET counts four of the top ten assembly and test firms by 2024 revenue as China-headquartered, and their equipment is far less tightly controlled than lithography or deposition.

WHERE THE US STANDS

Strong in test equipment and probe cards through Teradyne, FormFactor, Cohu and Aehr, weak in assembly capacity. Amkor is building a US$7 billion packaging and test campus in Arizona and has a long-term partnership with TSMC to fill it.

Semiconductor manufacturing runs to 400 to 600 steps, and Advantest counts test as the only one that puts electricity through the chip [1]. Everything before this chapter makes dies; the back end of the industry, assembly and test, decides which of them are allowed into a package.

How it works

Each round of test catches failures the round before it cannot see.

- **Wafer sort.** Thousands of needles land on each die while it is still on the wafer, a tester drives patterns in and checks the answers, and the failures go no further.
- **Burn-in.** Run the part hot and under power for hours, so anything that would fail early fails now, before assembly.
- **Final test.** The tester runs the packaged part again, because packaging adds failures of its own.
- **System-level test.** The finished part sits in a socket and runs something close to a real workload, because some failures appear only when everything runs at once.

What changed is the value of proving a die good before it is packaged. Aehr screens for early and hidden failures before dies are cut from the wafer, so a weak processor never reaches high-bandwidth memory and an expensive substrate [2]. Its lead AI customer is moving burn-in off the finished system and onto the wafer, nine 300 mm wafers at a time [3].

Variants and trade-offs

Wafer sort and known-good die

The probe card that holds those needles is the hard part. For a high-bandwidth memory stack it must contact eight, twelve or sixteen dies at speed without degrading the signals [4]. For a large logic die every needle must sit in the same

plane, survive repeated touchdowns without drifting, and do it hot. Cards are custom to the device, so each new part means a new card and a new qualification.

Final test, burn-in and system-level test

The economics get worse as packages grow. TSMC has told investors that its CoWoS packaging roadmap runs beyond 14 times the area a lithography machine can print in one shot [5]. Every added die is another chance to lose the whole assembly, which is why the number of test steps keeps rising. ASE ran 6,797 testers in the second quarter of 2025 and 8,348 a year later [6].

Automated test equipment: share of the ~US$9.0bn tester market, 2025 %



Advantest **65%** Teradyne and all others **35%**

Source: Advantest FY2025 results briefing, April 2026

Who makes it

Advantest puts its own share at about 65% of a US$9.0 billion tester market in 2025: 66% of test for logic chips, up ten points in a year, and about 60% of memory test [7]. It and Teradyne hold about 80% between them [1]. Teradyne grew 13% to US$3.19 billion for 2025, with fourth-quarter revenue up 44% on AI demand [8].

Where the tester money went in calendar 2025 US$BN



SoC testers **6.9** Memory testers **2.1**

Source: Advantest FY2025 results briefing, April 2026

Below the two tester makers, the floor breaks into niches with their own suppliers.

- **Probe cards.** FormFactor took US$785.0 million in fiscal 2025 [9]; Technoprobe grew 15.7% to EUR 628.4 million, 38% of it from AI, and plans to double capacity by the end of 2027 [10].

- **Handlers and burn-in**, the machines that feed parts into a tester and cook them. Cohu took US$453.0 million in fiscal 2025 [11]; Aehr did US$50.0 million in fiscal 2026 but booked a record US$60.7 million in the fourth quarter alone and guides to US$130 million to US$150 million in fiscal 2027 [3].
- **Dicing and grinding.** Disco leads in wafer grinders and the saws that cut wafers into dies [12], and its newest saw handles pieces up to 400 by 400 mm, for packaging on rectangular panels rather than round wafers [13].
- **Bonding.** Kulicke and Soffa did US$654.1 million in fiscal 2025 [14], while Besi's second-quarter 2026 revenue rose 68.7% to EUR 249.9 million on hybrid bonding and datacentre demand [15].

Thermocompression bonders, which press memory dies into a stack and place chiplets on substrates, are the tightest niche. ASMPT took a repeat order for fifteen chip-to-substrate bonders in December 2025 and expects that market to pass US$1 billion by 2027, of which it means to hold 35 to 40% [16]. See Memory and HBM.

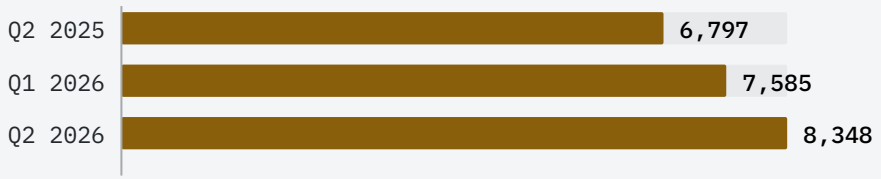
The back end's geography

Outsourced assembly and test, the OSAT business, is Taiwanese with a large Chinese challenger. ASE, the biggest, billed US$20.6 billion in 2025 [17]; Georgetown's CSET counts four of the top ten OSATs by 2024 revenue as China-headquartered [18]. Malaysia ships about 13% of the world's packaged chips and wants 7% of advanced packaging shipments by 2035 [19].

ASE spent US$2.7 billion on machinery and US$1.4 billion on buildings in the first half of 2026, and says revenue from leading-edge advanced packaging is running ahead of its US$3.5 billion guidance for the year [20]. Powertech is chasing the first panel-level packaging line for AI chips in 2027 [21].

Amkor took US$6.71 billion of revenue in 2025 [22], and US$1.90 billion in the second quarter of 2026 alone, up 26% year on year [23]. It has expanded its Arizona campus to a US$7 billion investment [24]. Under a partnership signed in June 2026 [25] it will run packaging and test there for TSMC, whose Phoenix fabs sit next door [26].

Testers installed at ASE, the largest assembly and test contractor TESTERS



Source: ASE Technology Holding, second quarter 2026 earnings release (SEC Form 6-K)

The chokepoint

Unlike lithography, no tool here has a single supplier. What holds a customer is the cost of leaving. Advantest argues that switching tester vendor means rebuilding the whole development, evaluation and production test environment, which is why share moves slowly [1].

This is also the part of the chain China is closest to owning, and assembly, dicing, bonding and test tools sit outside the export control lines drawn around lithography and deposition. ASE and Amkor billed US$27.3 billion between them in 2025, against industry sales of US$791.7 billion [27], a small share of the money for a step every chip has to pass through.

Key evaluation criteria

- **Test coverage** is what you buy: the share of hidden failures caught before they reach a package that cannot be taken apart.
- **Test time per part** sets throughput and cost, and large dies, memory stacks and chiplets all push it up.
- **Probe card flatness and life** decide what wafer sort costs. Every needle must stay in the same plane across a full die and hold that over a long life.
- **Thermal control** matters more every generation. A kilowatt-class accelerator has to be held at temperature in the socket while it is tested at speed.
- **Capacity and geography** are the practical constraint. Taiwan, China and Malaysia hold most of the floor space, and new capacity takes years.

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CHAPTER
14
STAGE 14 OF 15

Systems and Networking

A finished package is useless until it is bolted to a baseboard, fed a thousand amps and wired to seventy-one siblings. Many firms can do the bolting; few can make the lasers.

IN PLAIN TERMS

A rack is where finished chips become one machine. Builders bolt each package onto a board, feed it more than a thousand amps, and wire it to seventy-one neighbours so that a training run sees one computer instead of seventy-two. A cluster runs only as fast as its slowest link, and copper carries a signal a few metres at most, so the longer links have to become light. Many firms can bolt racks together. Far fewer can make the lasers that carry the light, and that is where the chain narrows.

IN SHORT

The system layer is where AI hardware stops being a chip business and becomes an electrical and optical one. Assembly is competitive and can move countries in a year. The control points are NVLink-class scale-up switching, which one firm still owns [3], and indium phosphide lasers, where Nvidia paid $4 billion for capacity it could not order [17] and 70% of the feedstock is Chinese [16].

CHOKEPOINT CARD ●●●○○ 3 / 5

CONCENTRATION

High

SUBSTITUTION TIME

2-3 years

PRICE OR MARKET SIZE

Nvidia put $2B each into Coherent and Lumentum in March 2026 to buy indium phosphide laser capacity

WHO LEADS

- US **Nvidia** NVLink and NVSwitch for links inside a rack; InfiniBand and Spectrum-X Ethernet between racks
- US **Broadcom** Switch chips sold to all comers (Tomahawk, Jericho) and most custom AI accelerators
- TW **Foxconn** Largest AI rack assembler; builds in Taiwan, Mexico, Texas
- CN **InnoLight** High-speed optical modules; plants in Suzhou, Taiwan and Thailand
- US **Coherent** Indium phosphide lasers and data centre transceivers

WHERE IT IS MADE

- US **United States** Switch and accelerator silicon, indium phosphide lasers, system design
- TW **Taiwan** Contract design and rack assembly; high-layer-count circuit boards
- CN **China** Optical module assembly and test; most of the world's indium
- TH **Thailand** Chinese module makers' offshore transceiver plants
- MX **Mexico** Rack integration for the North American market

WHY SUBSTITUTION IS SLOW

Rack assembly and switch boxes have many credible builders, but indium phosphide laser fabs take 18 to 24 months to ramp and there are only a handful of them.

WHERE CHINA STANDS

Holds the two ends of the optical link that the United States does not. Chinese firms assemble high-speed modules at scale, mostly from Thai and Suzhou plants, and China accounts for about 70% of world indium production, the feedstock for the lasers inside them.

WHERE THE US STANDS

Owns the switch silicon, the protocols that link accelerators inside a rack, and the laser fabs, but almost none of the rack assembly. Nvidia's partners began US rack production in Houston and Dallas in 2025, and the Federal Communications Commission moved in July 2026 against imported gear containing parts from firms on its Covered List.

A package computes nothing until it is wired into a machine, and bolting it down, powering it and linking it to its neighbours is a separate industry from making it.

How it works

An accelerator ships as a module, not a bare chip: Nvidia calls its version SXM, and the open-standards equivalent is the Open Compute Project's accelerator module. Eight bolt onto a baseboard that carries their power and the switch chips that let them reach each other at full speed.

Rack-scale systems do away with the baseboard. An Nvidia GB300 NVL72 holds 18 compute trays plus nine switch trays, 72 GPUs in all, and draws up to 142 kW [1]; Rubin keeps that shape [2].

Every AI cluster runs two networks that are not interchangeable. Scale-up binds a few dozen accelerators tightly enough that they can read each other's memory, at terabytes per second; scale-out joins thousands of those groups at hundreds of gigabits per second. A model too big for one accelerator gets split across the scale-up group, so that group's bandwidth sets the size of model you can train.

Variants and trade-offs

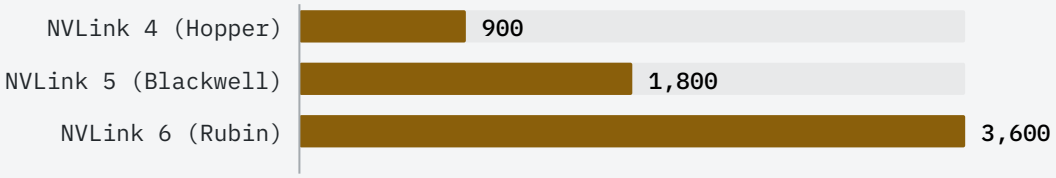
Scale-up: NVLink, UALink and Google's torus

The designs differ in how many accelerators can share one pool of memory.

- **NVLink and NVSwitch.** Nvidia's per-GPU bandwidth went from 900 GB/s in NVLink 4 to 1,800 in NVLink 5 and 3,600 in NVLink 6; the rack's own network moves 130 TB/s on Blackwell and 260 TB/s on Rubin [3].
- **UALink.** The open answer runs 200 Gb/s per lane, four lanes to a station for 800 Gb/s, and can address up to 1,024 endpoints in one group [4].

- **Google's 3D torus.** Ironwood, the seventh generation of Google's own TPU accelerator, wires 64 chips to a rack and uses optical switches to stitch cubes into superpods of 9,216 chips, trading latency for the ability to route around a failed rack in software [5].

NVLink bandwidth per GPU by generation GB/S

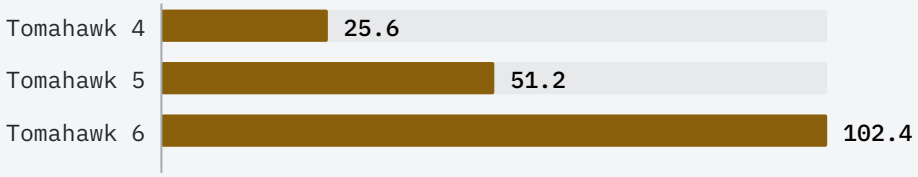


Source: NVIDIA NVLink product page, 2026

Scale-out: InfiniBand versus Ethernet

InfiniBand, the specialised network supercomputers grew up on, owned this layer when AI clusters were small and alike. Ethernet took it back on raw capacity first: Broadcom doubled the bandwidth of the switch chips it sells to all comers every generation, from 25.6 Tbps on one chip in Tomahawk 4 [6] to 51.2 in Tomahawk 5 [7] and 102.4 in Tomahawk 6 [8]. Then the Ultra Ethernet Consortium published its 1.0 specification in June 2025, standardising the load balancing, congestion control and lossless behaviour InfiniBand had from the start [9].

Broadcom Tomahawk switching capacity on one chip TBPS



Source: Broadcom product releases for Tomahawk 4, 5 and 6

Nvidia sells both sides of the argument, telling investors that a $75.2 billion data center quarter came from Blackwell 300 plus demand for "InfiniBand, Spectrum-X Ethernet, and NVLink" [10], because large buyers run both networks.

Optics: pluggable, then co-packaged

Every link longer than a few metres is optical, and a cluster needs several transceivers, the modules that turn electricity into light and back, for each accelerator. Their power cost is why both switch vendors are moving the optics

onto the switch package itself: Broadcom's Tomahawk 6 Davisson cuts optical interconnect power by about 70%, more than 3.5 times better than plug-in modules [11], and Nvidia's Spectrum-X Photonics reaches 512 ports of 800 gigabits during 2026 using four times fewer lasers [12].

Who makes it

Optical modules are assembled where labour and land are cheap, and the Chinese makers went offshore early. InnoLight has run a Thai plant since 2019 alongside Suzhou and Taiwan [13], and Eoptolink assembles in Rayong and Chonburi [14]. The rules now follow the firm rather than the site: the Federal Communications Commission's July 2026 order bars new authorisations for devices that contain logic chips from firms on its Covered List, and the record for that rule names optical transceivers as exactly such a component [15].

The lasers inside those modules come from far fewer hands. Indium phosphide is the only practical material for the lasers in 800-gigabit and 1.6-terabit optics, and its feedstock is concentrated: China accounts for about 70% of world indium production [16]. Nvidia put $2 billion into Lumentum and $2 billion into Coherent in March 2026, buying research, future capacity and access rights rather than parts [17] [18].

Custom accelerator and switch silicon is a US duopoly. Broadcom sells the Ethernet switch chips other firms build boxes around, and designs most of the custom accelerators, including Google's TPUs and Meta's own; its AI semiconductor revenue reached $16.7 billion in the third quarter of fiscal 2026, up 221% [19]. The other custom-silicon designer is Marvell, whose five-year agreement with AWS, announced in December 2024, covers custom AI chips alongside the optical and networking parts that go with them [20].

Physical assembly belongs to Taiwan. Foxconn booked NT$8.1 trillion of revenue in 2025, up 18%, with cloud and networking among the drivers [21]. Assembly is moving closer to the customer: Nvidia's partners commissioned more than a million square feet of US manufacturing space in 2025, Foxconn in Houston and Wistron in Dallas [22].

The chokepoint

Most of this layer is not a chokepoint. Several capable contract manufacturers on three continents screw racks together, and the power shelves, copper busbars,

cables and connectors have many suppliers. If Foxconn stopped tomorrow, Quanta and Wistron would absorb the volume in a couple of quarters.

What cannot be replaced that fast sits underneath the assembly.

- **Indium phosphide laser capacity.** The shortage is in material and fab space: Nvidia's answer was to buy into two laser makers rather than place a bigger order [18], and the indium under it is 70% Chinese [16].
- **NVSwitch.** Nobody else sells a scale-up switch of NVLink's bandwidth to all comers, which is what UALink exists to change [4].
- **High-layer-count circuit boards.** The boards, and the low-loss laminate they are built from, are concentrated in Taiwan and Japan; see Substrates and PCBs.

Optical module assembly is a separate case: concentrated, but not technically hard. Where the modules get built follows cost and policy, and it can change in a year or two.

Key evaluation criteria

- **Scale-up domain size** – how many accelerators share one pool of memory. Up to 1,024 endpoints in the UALink spec [4] and 9,216 for an Ironwood superpod [5].
- **Bandwidth per accelerator** – the number that limits how far one model can be split across chips. NVLink 6 is 3,600 GB/s [3].
- **Picojoules per bit**, the energy cost of moving data – optics dominate network power at 800 gigabits and above, and co-packaging is worth more than 3.5 times [11].
- **Serviceability** – co-packaged optics trade field-replaceable modules for a switch you return whole when a channel fails.
- **Supply concentration** – many firms assemble the modules; far fewer can make the lasers inside them.

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CHAPTER
15
STAGE 15 OF 15

Data Centers and Power

The chips are no longer what a buildout waits for. A large transformer takes three years, turbine output does not reach 30 GW a year until 2030, and Texas has fifty times more large load queued than it has approved to switch on.

IN PLAIN TERMS

A data center's job is to feed and cool the machines inside it. The scale has changed: a hair dryer pours about two kilowatts of heat out of a hand-sized nozzle, and one rack of AI chips draws seventy of them in a box the size of a wardrobe, every watt of it turning into heat that has to be carried away in water rather than air. The building is the easy part. Getting the electricity to the door is the hard one, because the transformers, switchgear and turbines that connect a site to the grid come from a few suppliers and are ordered years in advance.

IN SHORT

The rack became the unit of AI compute at 142 kW and is heading for a megawatt by 2027 [4], which forces liquid cooling and an 800-volt bus on every serious operator [3]. Nine vendors sell the cooling equipment, so it can be bought and haggled over [3]. The power equipment comes with queues: transformers take three years [1], six years of turbine output is already booked [9], and Texas has fifty times more large load in the queue than it has approved to switch on [13]. When an AI buildout slips, the delay is usually in the substation yard rather than the fab.

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

High

SUBSTITUTION TIME

4-6 years

PRICE OR MARKET SIZE

Transformer lead times of 36 months, against under a year before 2020

GE Vernova's gas backlog and reserved slots grew from 100 GW to 116 GW in 2026

WHO LEADS

- US **GE Vernova** 116 GW of gas equipment backlog and reserved factory slots at mid-2026
- CH **Hitachi Energy** Large power transformers and high-voltage switchgear
- DE **Siemens Energy** Grid technologies, transformers, turbines
- US **Vertiv** Data center power and cooling systems; NVIDIA gigawatt AI factory partner
- FR **Schneider Electric** Electrical distribution and cooling; partner in NVIDIA's 800-volt DC designs

WHERE IT IS MADE

- US **United States** Most new AI capacity; ERCOT and PJM carry the load growth
- CH **Switzerland** Hitachi Energy and ABB transformer and switchgear engineering
- DE **Germany** Siemens Energy grid technologies and turbines
- TW **Taiwan** Delta Electronics power shelves, busbars and thermal modules
- JP **Japan** Hitachi group transformer manufacturing

WHY SUBSTITUTION IS SLOW

Large power transformer lead times of 36 months are commonly quoted and can reach 60, GE Vernova is building toward 30 GW of annual turbine output only by 2030, and the one reactor restart with a data center buyer behind it is still waiting on its licence.

WHERE CHINA STANDS

Power is not China's binding constraint. It has surplus generation, a large domestic transformer and switchgear industry, and can energise sites quickly. Its AI buildout is limited by accelerator supply, not megawatts.

WHERE THE US STANDS

Has the demand and the capital but cannot shorten the equipment queues. Transformers and turbines are both multi-year waits, most large power transformers are imported, and no restarted reactor is running yet.

For thirty years the semiconductor industry set the pace and everything downstream followed. That has flipped. A wafer clears a leading-edge fab in months; a large power transformer takes three years, and before 2020 the same order took under one [1].

How it works

A rack is now the unit of AI compute. A GB300 NVL72 draws up to 142 kW through eight 33 kW power shelves, more than ten times a conventional enterprise rack in the same floor area [2].

Everything else follows from that number. Utility power steps down through a substation, then through switchgear, battery-backed supplies and distribution boards, before it reaches the busbar at the back of the rack. Each stage has its own lead time.

Heat leaves in water. Cold plates on the GPU packages pass it to a building loop that dumps it outdoors, and Nvidia's Vera Rubin racks run 45 °C supply water, warm enough that many sites can skip chillers [3].

Nvidia is also moving rack distribution to 800 volts of direct current, which pushes more than 150% more power through the same copper and retires the 200 kg busbars feeding a single rack [3].

Variants and trade-offs

Cooling

- **Air.** Heat exchangers in the rack door work at moderate densities but stop paying off well below AI rack loads.
- **Direct-to-chip liquid.** The default for anything Blackwell-class or later; every serious AI hall is now plumbed.
- **Immersion.** Higher density still, but hard to service and barely used at scale.

Cooling is not the bottleneck. Nvidia names nine firms supplying data center power and cooling systems for its gigawatt reference designs, among them ABB, Eaton, Hitachi Energy, Mitsubishi Electric, Schneider Electric, Siemens and Vertiv [3]. With nine suppliers, a buyer can get the kit and haggle over the price.

Rack power

Nvidia's 800-volt DC architecture is drawn for racks from 100 kW to more than a megawatt, with megawatt racks from 2027, because the 54-volt distribution in today's racks hits physical limits past about 200 kW [4].

A megawatt rack changes the busbar, the floor loading, the coolant flow, the leak detection and the stored energy that carries the rack through a dip: Nvidia's Rubin racks hold 20 times more of it, to steady the draw as tens of thousands of GPUs enter and leave a training step together [3].

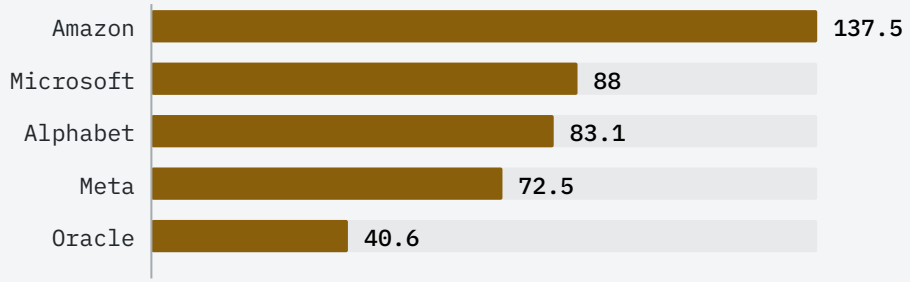
Where the electrons come from

- **Grid interconnection.** The slow path, and the one every announced campus is queued in.
- **Behind-the-meter gas**, generation on the site itself. The fast path, which is why campuses are sited next to gas rather than next to fibre.
- **Nuclear.** Almost entirely prospective. Constellation sold the 835 MW Crane Clean Energy Center forward to Microsoft before a spade went in [5]. The plant shut in 2019 and is restarting on a $1 billion federal loan closed in November 2025, still subject to Nuclear Regulatory Commission approval [6]. No restarted reactor is serving data centers yet.

Who makes it

The largest data center operators pay for it. Alphabet, Amazon, Meta, Microsoft and Oracle spent $448.3 billion of capital in 2025, and on that trend reach $770 billion in 2026 [7].

Capital spending by the five largest US data center operators, 2025 $B

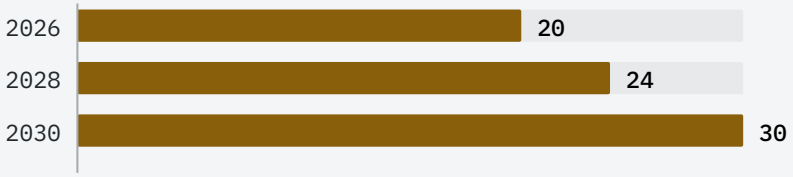


Source: Epoch AI, hyperscaler capex trend, February 2026, from company filings

Campuses are now described in gigawatts: OpenAI's Stargate portfolio targets more than 9 GW across US sites by 2029, with Abilene, Texas going from 0.3 GW to 1.2 GW by late 2026 [8].

The equipment vendors are the constraint. GE Vernova, Siemens Energy, Hitachi Energy, ABB, Eaton and Schneider Electric supply the turbines, transformers and switchgear. GE Vernova's gas equipment backlog and reserved factory slots went from 100 GW to 116 GW inside 2026, with data center orders past $5 billion by mid-year, more than double the whole of 2025 [9].

GE Vernova annual gas turbine output, actual and planned GW/YR

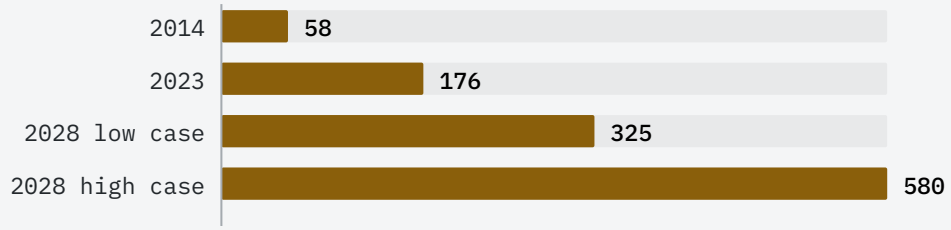


Source: GE Vernova second-quarter 2026 results, July 2026

Capacity and demand

US data centers used 176 TWh in 2023, 4.4% of national electricity, and Lawrence Berkeley National Laboratory projects 325 to 580 TWh by 2028, between 6.7% and 12% of the total [10]. The Energy Information Administration expects national electricity use to rise 1% in 2026 and 3% in 2027, the fourth straight annual increase and the strongest four-year run since 2000 [11].

US data center electricity use, actual and 2028 projection range TWH



Source: Lawrence Berkeley National Laboratory, December 2024

Chip efficiency keeps the low case plausible: Google's Ironwood TPU delivers twice the performance per watt of its predecessor [12]. The energy per token, the chunk of text a model reads or writes, falls while total energy rises, because the number of tokens rises faster.

The chokepoint

Announced capacity and energised capacity are a long way apart. ERCOT, the grid operator for most of Texas, was tracking about 474 GW of large loads seeking a connection in June 2026, roughly 90% of it data centers [13]. Only 9,042 MW had been approved to switch on [14].

Behind that queue sit the equipment lead times.

- **Large power transformers.** Thirty-six-month lead times are commonly quoted and the worst reach 60, against under a year before the pandemic; 82% of the units put into US service in 2019 were imported [1].
- **Gas turbines.** GE Vernova will build 20 GW of turbines this year and is working toward 30 GW a year in 2030, against a book of 116 GW, close to six years of output [9].
- **Firm new generation**, the kind that runs whatever the weather. The one reactor restart with a data center buyer behind it, the 835 MW Crane plant, shut in 2019 and is still waiting on its licence [6].

Set that against the chip chain, where foundry and memory capacity have repeatedly expanded within a couple of years of the order. Money buys silicon capacity in quarters and power capacity in years, and that difference decides how much of the 2026 capital spending turns into working compute.

Key evaluation criteria

- **Time to power** – months from signed lease to energised megawatt. This is the number developers compete on.
- **Transformer and turbine lead time** – the two queues that set the schedule, at three years [1] and six years of booked output [9].
- **Rack density supported** – an air-cooled hall cannot host a 142 kW NVL72 without a full mechanical retrofit [2].
- **Firmness of supply** – a signed interconnection agreement and a delivered transformer are different assets.
- **Energy per token** – the only efficiency measure that matters commercially. It keeps falling, but the number of tokens rises faster [12].

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CHAPTER
16
THE COSTS

Component Costs

Follow the money through an AI accelerator and the silicon turns out to be the cheap part. Memory and packaging are the bill.

IN PLAIN TERMS

A bill of materials is the list of parts inside a product and what each one costs to build. For an AI accelerator, the famous logic chip is the cheap part, about a seventh of the total. The stacked memory beside it is nearly half, and the packaging that wires the two together costs more than the chip. Three firms in the world make that memory and one Taiwanese firm does that packaging, so the costliest parts are also the scarcest, and that is where a government has the most purchase.

IN SHORT

On a B200 the logic dies are 14 percent of the build, memory 45, and packaging with its scrap another 33 [1]. Nvidia's 75 percent gross margin is the most visible number in the chain [6], and a custom chip can attack it by buying the same wafers and the same memory. The lines underneath are harder to attack: memory stacking yield, CoWoS interposer yield and the megawatts do not get cheaper as volume rises, which is why we treat them as chokepoints.

CHOKEPOINT CARD ●●●●○ 4 / 5

CONCENTRATION

High

SUBSTITUTION TIME

3-5 years

PRICE OR MARKET SIZE

~$6,400

of manufacturing cost inside a B200 that sells for $30,000 to $40,000. memory alone is $2,900 of it

WHO LEADS

- US **Nvidia** 75.0% gross margin in the July 2026 quarter; $193.7B data centre revenue in FY2026
- US **Micron** 84.6% gross margin in the May 2026 quarter, up from 39.8% for FY2025
- TW **TSMC** 67.7% gross margin in the June 2026 quarter, 59.9% for 2025
- TW **ASE Technology** 21.0% gross margin, the thin end of the chain

WHERE IT IS MADE

- US **United States** Accelerator design and the largest single pool of profit
- TW **Taiwan** Wafers, CoWoS packaging, outsourced assembly and test, and server build
- KR **South Korea** High-bandwidth memory, the single largest line on the bill
- JP **Japan** Ajinomoto build-up film and the substrates made from it
- NL **Netherlands** Lithography, the largest single line in fab capital cost

WHY SUBSTITUTION IS SLOW

A custom chip built on the same TSMC wafers and the same memory can attack Nvidia's margin. The memory and packaging lines underneath cannot be attacked that way, because physically scarce capacity sets them.

WHERE CHINA STANDS

Domestic accelerators dodge Nvidia's margin and pay for it elsewhere: no advanced packaging at volume, memory bought from stockpiles or from CXMT's early product, and yields on SMIC's N+2 process that make the die the expensive part rather than the cheap one.

WHERE THE US STANDS

Captures the design and equipment margins and almost none of the wafer, memory, substrate or assembly cost, which is why its export controls target access rather than production.

Building a Blackwell B200 costs about $6,400. Nvidia sells it for $30,000 to $40,000 [1]. Follow that $6,400 and you learn which stages are expensive because of physics, which because capacity is short, and which because one firm can charge.

How it works

Three numbers set the cost of a die, the finished rectangle of circuitry cut out of a wafer: the price of the wafer, how many dies fit on it, and how many of those work.

- **Wafer price.** TSMC does not publish it. Epoch AI's model of Blackwell's custom TSMC 4NP process [2] assumes about $17,000 for a 300 mm wafer [1].
- **Dies per wafer.** A wafer is a circle and dies are rectangles, so the edge of the disc goes to waste: an 800 mm2 die, near the largest a scanner can print in one shot, fits 68 times.
- **Yield.** Defects land at random, so a bigger die is likelier to catch one. Epoch AI models the Blackwell die at 40 to 70 percent good, centred on 60 [1].

Run those three numbers and the two dies inside a B200 come to about $900, 14 percent of the build [1]. Doubling the wafer price moves the whole bill by a few hundred dollars.

Variants and trade-offs

Memory is the bill

The 192 GB of high-bandwidth memory beside those dies, sold as HBM3E, is the single largest line: about $2,900, 45 percent of the build, at $14 to $17 per gigabyte [1]. Nvidia controls that line least of all: three firms make the memory, and they sell every stack they can build.

Packaging, substrate and test

CoWoS-L, the TSMC process that mounts dies and memory on one carrier and wires them together, adds about $1,100, more than the logic dies it carries [1]. Packaging yield runs 65 to 95 percent, and a package that fails after the memory goes on throws away the dies and the memory with it, adding roughly $1,000 to every B200 that ships [1]. Substrate, board, test and assembly add $480 [1]. See Advanced packaging for why that capacity binds first.

The B200 bill of materials

LINE	ESTIMATE	RANGE
HBM3E, 192 GB in eight stacks	$2,900	$2,800-3,100
CoWoS-L packaging	$1,100	$1,000-1,200
Packaging yield loss	$1,000	$430-1,700
Logic die, two on TSMC 4NP	$900	$720-1,200
Substrate, board, test, other	$480	$370-600
Manufacturing cost	~$6,400	$5,700-7,300
Selling price	$30,000-40,000	

Epoch AI models each line as a range rather than a single figure, deriving the packaging line from TSMC's advanced packaging revenue and Nvidia's share of CoWoS capacity [1]. Nothing of that quality exists for the H100 or Rubin, so the table carries one generation only.

Where the ~$6,400 of B200 manufacturing cost goes, 2026 $



HBM3E, 192 GB **2,900** CoWoS-L packaging **1,100** Packaging yield loss **1,000**

Logic die, two on 4NP **900** Substrate, board, test **480**

Source: Epoch AI B200 cost model, from a $17,000 4NP wafer, 40-70% die yield, $14-17 per GB of memory, and a packaging line derived from TSMC's advanced packaging revenue

From package to rack to cluster

Buy a whole rack and the number that matters is the cost of a GPU-hour. For an operator that owns the machine, SemiAnalysis puts a Vera Rubin NVL72 at $3.57, against $1.84 for a GB200 and $2.36 for a GB300 [3]. Each generation costs its owner more per hour than the last, and buyers pay because throughput rises faster than the rate.

One level up, a 100,000 H100 cluster costs over $4B in servers, draws about 150 MW, burns 1.59 TWh a year, pays $123.9M for power and needs 98,304 optical transceivers [4].

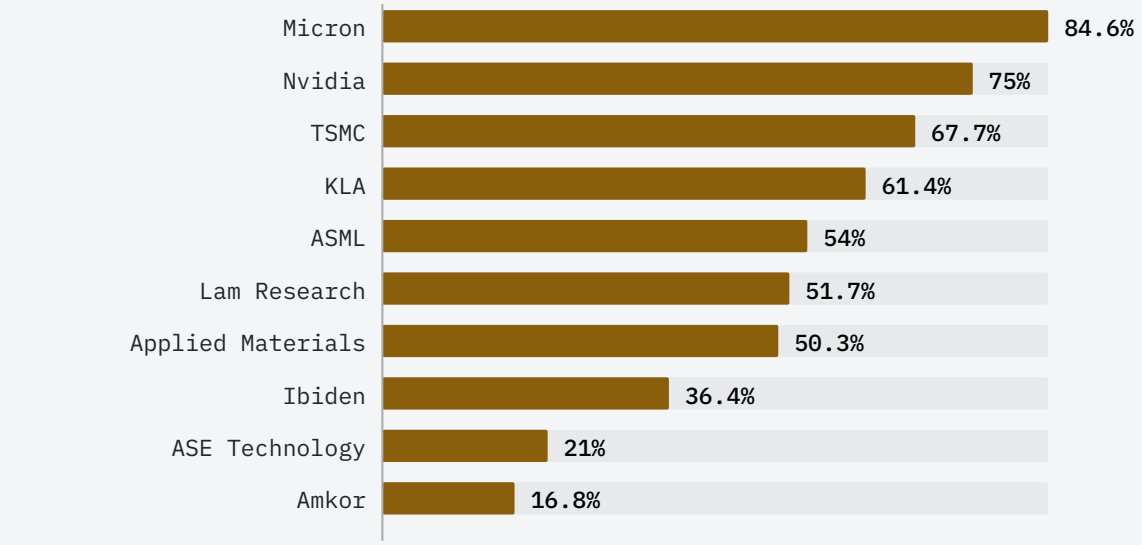
Who captures the value

FIRM	STAGE	QUARTER ENDED	Gross margin
Micron	HBM and DRAM	May 2026	84.6%
Nvidia	Accelerator design	July 2026	75.0%
TSMC	Wafers and CoWoS	June 2026	67.7%
KLA	Inspection	June 2026	61.4%
ASML	Lithography	June 2026	54.0%
Lam Research	Etch	June 2026	51.7%
Applied Materials	Deposition	August 2026	50.3%
Ibiden	Substrates	June 2026	36.4%
ASE Technology	OSAT	June 2026	21.0%
Amkor	OSAT	June 2026	16.8%

Every figure is the firm's own reported gross margin [5] [6] [7] [8] [9] [10] [11] [12] [13] [14].

SK hynix publishes no gross margin. Its operating margin hit 76 percent in the June quarter, on revenue up 257 percent in a year [15].

Gross margin, each firm's most recent reported quarter %

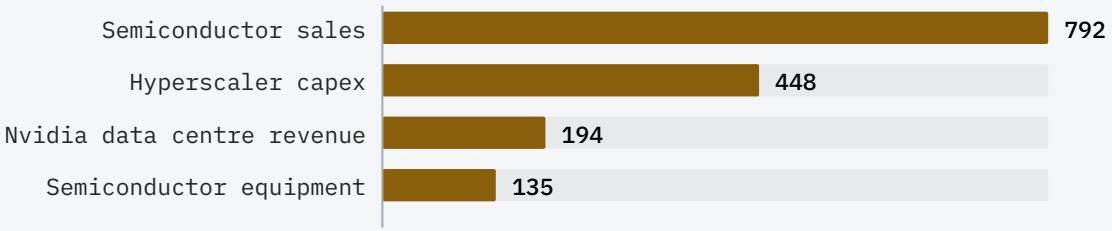


Source: Each firm's own quarterly earnings release; quarters end between May and August 2026 (TSMC shown)

Design, inspection and leading-edge wafers earn gross margins of 50 to 75 percent, substrates 36, assembly and test 17 to 21. In 2026 memory passed all of them: Micron earned 39.8 percent for the year to August 2025 [16] and 84.6 percent in the quarter to May 2026 [5]. The margin moved up the chain, to Nvidia's suppliers.

The two bodies that count this industry disagree, which sets the error bar here. World Semiconductor Trade Statistics forecast $772 billion of global chip sales for 2025 [17]; the Semiconductor Industry Association counted $791.7 billion [18]. The forecasters put 2026 at $975 billion, with memory and logic each growing more than 30 percent [17]. Below those totals sit $135.1B of equipment billings [19] and $448.3B of 2025 capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle, on track for $770B in 2026 [20].

Size of each layer of the chain, 2025 $B



Source: Semiconductor Industry Association on chip sales; Epoch AI on capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle; Nvidia FY2026 results; SEMI on equipment billings

The chokepoint

Per unit of compute, logic keeps getting cheaper and everything wrapped around it gets dearer. Packaging and its scrap cost more than twice the logic dies [1], and a GB300 NVL72 rack already draws 142 kW [21].

STAGE	CLASS	WHY
EUV lithography	Structurally expensive	One supplier; price set by a twenty-year optics curve.
Masks and pellicles	Structurally expensive	Scales with layer count; every design pays again.
Metrology and inspection	Structurally expensive	Inspection time rises faster than throughput.
HBM	Structurally expensive	Stacking and through-silicon vias cut good die per wafer.
Advanced packaging, CoWoS-L	Structurally expensive	65 to 95 percent packaging yield; scrap destroys dies and memory.
Power and cooling plant	Structurally expensive	Priced by grid queues and turbine lead times.
Leading-edge logic wafers	Compressible with scale	Defect density and utilisation improve through a ramp.
Design and EDA	Compressible with scale	Engineering cost amortises across volume.
Deposition and etch tools	Compressible with scale	Credible second sources in most steps.
ABF substrates	Compressible with scale	A laminate line, not a physics frontier.
Optical transceivers	Compressible with scale	Copper and co-packaged optics cap the price.
Test and assembly (OSAT)	Already commoditised	17 to 21 percent margins; clears on price.
Wafers and bulk gases	Already commoditised	Multi-supplier, long-qualified, thin spreads.

STAGE	CLASS	WHY
Rack integration (ODM)	Already commoditised	Low-teens margins on specified content.

The classification is ours; its yield and margin figures are cited above.

Key evaluation criteria

- **Cost per good die.** A 30 percent wafer price rise at constant yield is survivable. A ten-point yield drop on an 800 mm2 die is not.
- **Memory dollars per gigabyte, and gigabytes per part.** The largest line on the bill, set by three firms.
- **Packaging yield.** Every point lost throws away finished dies and the memory attached to them [1].
- **Share of the bill the vendor does not control.** Nvidia sets its own margin but buys memory, packaging and substrates at another firm's price.
- **Rack watts.** Past about 140 kW a rack, power and cooling capex grows faster than accelerator capex [21].
- **Compressibility class.** Does the line fall when volume doubles? If not, capacity is the constraint.

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CHAPTER
17
THE CONTEST

Geopolitics

Every stage in this chain sits in someone's jurisdiction. Since 2018 the United States has been converting that fact into policy, and China has been building an answer.

IN PLAIN TERMS

An export control is a licence requirement. A government names an item and a buyer, and its own companies then need a licence to ship that item to that buyer; Washington stretches the rule to cover foreign-made goods built with American technology. Few governments can enforce such a rule, because the machines, software and chemicals that make an advanced chip come from a short list of firms in the United States, the Netherlands, Japan and South Korea. China can copy the simpler machines within a few years. It cannot yet copy the machine that prints the finest circuits, the chemistry that machine needs, or high-bandwidth memory.

IN SHORT

One or two countries hold a dozen points in the chain, and export controls try to turn that into leverage. They have held China below the leading edge, where extreme-ultraviolet machines, their resist and high-bandwidth memory stay out of reach. They have not stopped substitution beneath it, where domestic tools now take a third of China's own market. The Netherlands and Japan paid for that, their toolmakers' China revenue falling by a third or more in two years. What began as a small yard is now tariffs, equity stakes and a cut of licensed sales, and none of it has moved leading-edge logic or packaging out of Taiwan.

CHOKEPOINT CARD ●●●●● 5 / 5

CONCENTRATION

Extreme

SUBSTITUTION TIME

10+ years

PRICE OR MARKET SIZE

$52.7B

authorised under the CHIPS Act against about $47B in the latest round of China's Big Fund alone

WHO LEADS

- US **United States** 93% of AI accelerator design revenue; 72% of measurement and inspection and 60% of deposition tool revenue
- TW **Taiwan** 78% of the world's contract chipmaking and 92% of capacity for the most advanced chips
- CN **China** about half of world capacity for mature-node and commodity chips; about 10% of deposition and etch tool revenue
- NL **Netherlands** 100% of extreme-ultraviolet and 95% of deep-ultraviolet lithography revenue
- KR **South Korea** SK hynix alone took 56% of high-bandwidth memory revenue and 29% of the wider DRAM market in 1Q26

WHERE IT IS MADE

- US **United States** Chip design, design software, three of five tool segments, and the export-control authority itself
- TW **Taiwan** Leading-edge logic and advanced packaging, and the concentration risk no policy has reduced
- CN **China** Mature-node capacity, critical minerals, and the target of the whole regime
- NL **Netherlands** ASML lithography, the hardest single chokepoint in the chain
- JP **Japan** Photoresist, silicon wafers, etch and clean tools, and Rapidus

WHY SUBSTITUTION IS SLOW

China can copy mature tools in three to five years. Extreme-ultraviolet optics, the resist those machines need and memory stacking each took two decades and a supplier base no national programme can build to order.

WHERE CHINA STANDS

Blocked at the leading edge and substituting fast below it. Domestic tools went from 25 percent of China's own equipment market in 2024 to 35 percent in 2025, and about 40 percent in etch and deposition; SMIC runs 7 nm-class logic without extreme-ultraviolet machines at roughly 45,000 advanced-node wafers a month; CXMT's 2026 memory output covers 250,000 to 300,000 Ascend packages. Neither those machines nor high-bandwidth memory at scale is in reach before 2028.

WHERE THE US STANDS

Holds chip design, design software and most process tools, and uses the Foreign Direct Product Rule to reach foreign-made goods. Has traded parts of the fence for revenue and investment since 2025: an expected 15 percent cut of licensed H20 sales, a 9.9 percent equity stake in Intel, a 25 percent tariff on advanced chips, and case-by-case H200 licences.

2,493 WORDS / 11 MIN

Washington's control regime and Beijing's answer to it are both unfinished. What follows is a status report on each.

The shape of the contest

In 2022 Washington changed the goal from staying a generation ahead to "as large a lead as possible", sold as a small yard with a high fence [1]. The October rules that followed aimed to push China backwards rather than merely slow it [2]. Licensing once asked who was buying and why; country-wide prohibition asks only where the box is going.

Chokepoints and who holds them

SEGMENT	LEADER	SHARE	BASIS
EUV lithography	Netherlands	100%	revenue, 2025

SEGMENT	LEADER	SHARE	BASIS
DUV lithography	Netherlands	95%	revenue, 2025
Metrology and inspection	United States	72%	revenue, 2025
Deposition	United States	60%	revenue, 2025
Etch and clean	United States	53%	revenue, 2025
Photoresist	Japan	75%	revenue, 2024
Silicon wafers	Japan	51%	revenue, 2025
The most advanced logic	Taiwan	92%	capacity, 2025
Foundry, all nodes	Taiwan	78%	market share, 2025
HBM	South Korea	56%	SK hynix alone, revenue, 1Q26
AI accelerator design	United States	93%	revenue, 2025
Mature-node and commodity chips	China	about half	capacity, 2025

- **Equipment.** CSET's ETO Supply Chain Explorer, TechInsights 2025 revenue by parent-company headquarters [3].
- **Materials.** CSET's 2021 segment map [4].
- **Taiwan.** The Stimson Center [5].
- **Memory.** SK hynix's US listing prospectus gives its first-quarter 2026 shares: 56.4 percent of high-bandwidth memory revenue and 29.1 percent of the wider DRAM market [6]. Samsung holds most of the rest, and no source we trust splits it.
- **Design and mature nodes.** CSIS [7].

Advanced packaging belongs in that table, but nobody publishes a capacity split and TSMC will not give one. No country in the table is self-sufficient, so controls bite in both directions.

The US export-control campaign, 2018 to today

Commerce built the regime one rule at a time, each closing a gap the last one left.

- **17 August 2020, the Foreign Direct Product Rule.** The Entity List, Commerce's roster of buyers that need a licence, cut Huawei and 68 affiliates off from US technology in May 2019 [8], so Huawei bought foreign chips instead. Commerce then extended the rule to any foreign item made with US technology and bound for Huawei [9]. That reach is what makes American controls extraterritorial.
- **7 October 2022, the break.** Two new export classifications, 3A090 and 3B090, capped exportable AI chips and set fab thresholds at 16 nm FinFET logic, 18 nm half-pitch DRAM and 128-layer NAND [2]. They also barred US persons from servicing advanced Chinese fabs, emptying them of American engineers within days [10].
- **17 October 2023, the density fix.** Chip designers had tuned interconnect speed to slip under the 2022 line while keeping the compute, so Commerce's Bureau of Industry and Security (BIS) dropped that parameter for two others: total processing performance and performance density. A data-centre chip now needs a licence at 4,800 points of that performance score, or at 1,600 with a density of 5.92 [11].
- **2 December 2024, the widest package.** 24 equipment types, three software tools, the first controls on high-bandwidth memory and 140 Entity List additions [12].
- **15 January 2025, AI Diffusion.** A rule sorting the world into three tiers of compute allocation [13], pulled on 13 May, two days before it bound [14]. No tiered replacement has appeared.

The H20 sequence shows the drift. On 9 April 2025 the government told Nvidia it needed a licence to ship the H20 to China, to the D:5 group of arms-embargoed countries, and to any company headquartered or ultimately owned there, so no Chinese buyer could route the order through a foreign subsidiary [15]. Nvidia took a $4.5 billion charge on stock it could no longer sell. Licences came back in August at a price: US officials, the filing says, "have expressed an expectation that the USG will receive 15% of the revenue generated from licensed H20 sales", though no regulation codifies it [16].

Early 2026 set the current settlement.

- **13 January 2026.** BIS moved the H200 and AMD MI325X to case-by-case review, with third-party testing and a cap at 50 percent of what the exporter ships to US customers [17].

- **15 January 2026.** A national-security proclamation under Section 232 put a 25 percent tariff on the same class of chips, then exempted almost every American use, from data centres to startups [18].
- **12 February 2026.** Applied Materials paid $252 million, the statutory maximum, over $126 million of ion implanters routed through Korea to an Entity-Listed Chinese customer [19].

Eight rule changes that shaped the regime

- 2019-05-16
  Huawei and 68 affiliates added to the Entity List
- 2022-10-07
  First country-wide controls on AI chips and fab tools
- 2023-10-17
  Performance density added; A800 and H800 captured
- 2024-12-02
  24 tool types, HBM controls, 140 entity listings
- 2025-01-15
  AI Diffusion rule published, three country tiers
- 2025-05-13
  AI Diffusion rescinded
- 2025-08-11
  US officials seek 15% of licensed H20 sales
- 2026-01-13
  H200 and MI325X moved to case-by-case review

Source: BIS press releases and Federal Register rules

Allies: the Netherlands, Japan, Korea and Europe

Unilateral tool controls would only transfer the sales, so the campaign depends on The Hague and Tokyo.

- **The Netherlands.** The Dutch began licensing advanced deep-ultraviolet tools on 1 September 2023, widened that in September 2024 [21] and reached measurement and inspection tools on 1 April 2025 [22].

- **Japan.** Controls on advanced manufacturing equipment took effect on 23 July 2023 and apply to all destinations rather than naming China [23].
- **South Korea.** BIS revoked the Validated End-User status that let Samsung, SK hynix and Intel run their China fabs without individual licences, effective 31 December 2025. It will license those fabs to keep running, but not to expand or upgrade [24].

On 30 September 2025 a BIS rule extended Entity List restrictions to any subsidiary half-owned by a listed firm [25]. The same day the Dutch economy minister invoked the Goods Availability Act against Nexperia, the Nijmegen chipmaker that supplies European car lines, citing governance failures that risked moving technology out of Europe [26]. Neither lasted: BIS suspended its rule for a year on 10 November [27] and the minister his order on 19 November [28]. Europe holds a chokepoint at the leading edge and none at the mature nodes its car industry runs on.

The MATCH Act, introduced on 8 April 2026, would ban sales of immersion deep-ultraviolet scanners to China, name CXMT, Hua Hong, Huawei, SMIC and YMTC in statute, and give allies 150 days to match before the US extends the Foreign Direct Product Rule alone [29].

China's response

China's retaliation goes at the inputs rather than the chips. Micron was first, failing a Cyberspace Administration security review on 21 May 2023 and losing its critical infrastructure buyers [30]. Mineral controls followed.

- **3 December 2024.** Notice 46, from China's Ministry of Commerce, banned exports of gallium, germanium, antimony and superhard materials to the United States, one day after the American chip rules [31].
- **4 April 2025.** Announcement 18 put seven medium and heavy rare earths under licensing, from samarium to yttrium, as metal, oxide, compound or finished magnet [32].
- **9 October 2025.** A further announcement reached foreign-made goods containing Chinese rare earths worth 0.1 percent or more of their value, putting a Chinese export licence between a foreign firm and its own product [33].
- **November 2025.** After Trump and Xi met at the Asia-Pacific summit, the ministry suspended the gallium ban for a year [34].

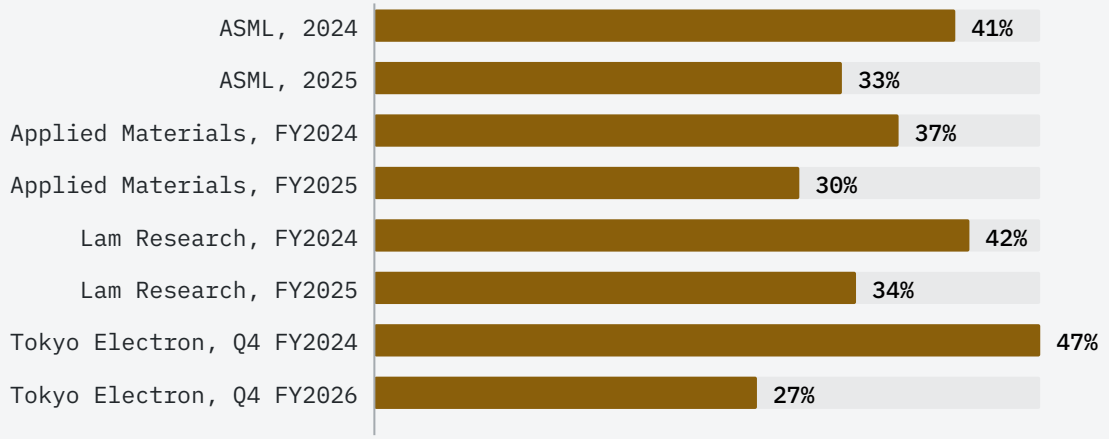
Nothing was repealed; the licensing regimes stand.

The blunter lever is demand. Chinese regulators discouraged customers from buying, importing or using Nvidia's data-centre products, and endorsed performance-per-watt and bandwidth standards for accelerators in Chinese data centres [35]. When Washington licensed H200 sales in February 2026, Beijing restricted them, and Nvidia's shipments came to less than 1 percent of data-centre revenue [35]. That quarter's was $89.0 billion, and the next quarter's guidance assumed no China compute revenue [36].

Chinese tool vendors went from 10 to 15 percent of their home market before 2022 to 25 percent in 2024 and 35 percent in 2025, with about 40 percent of domestic etch and deposition [7].

The leading edge is where it stalls. SMIC prints 7 nm-class logic on older deep-ultraviolet scanners, exposing each layer several times because it cannot buy the finer light, and Huawei shipped about 805,000 Ascend units in 2025. The binding constraint is memory. A stockpile of roughly 13 million memory stacks, most bought before the December 2024 rules bit, supports about 1.6 million Ascend 910C packages; CXMT, China's own memory maker, should reach about 2 million stacks in 2026, enough for 250,000 to 300,000 more [37].

China's share of revenue at the four largest chip-tool makers, peak year and latest %



Source: ASML Q4 2025 investor presentation, Applied Materials and Lam Research 10-K filings, Tokyo Electron Q4 FY2026 results

Industrial policy: CHIPS, Rapidus, Big Fund and the rest

The CHIPS and Science Act authorised about $52.7 billion [7]. By July 2025 $30.9 billion of direct funding had reached 19 companies for 40 projects, a median 14.2 percent of each project's capital cost [39].

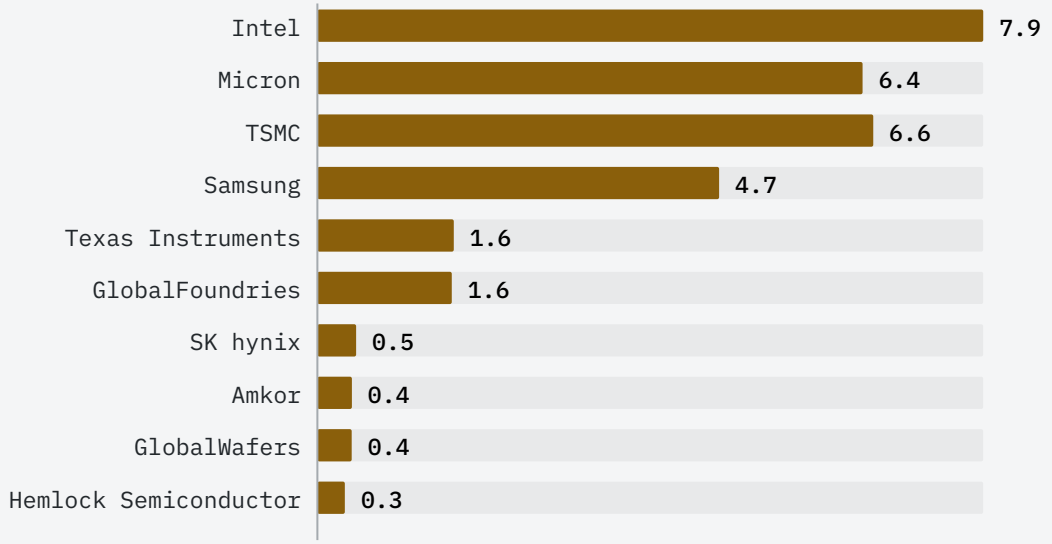
On 22 August 2025 the government converted $5.7 billion of unpaid Intel grants and $3.2 billion of Secure Enclave defence money into a 9.9 percent stake worth $8.9 billion [40].

Tariffs became the second instrument. The 2026 US-Taiwan trade agreement cuts the tariff on Taiwanese goods to 15 percent in exchange for at least $250 billion of Taiwanese investment in US chip production [41]. On 16 July 2026 TSMC added $100 billion for four more Arizona fabs at 2 nm and below, taking that campus to $265 billion [42], more than the whole CHIPS Act authorisation.

Europe's first Chips Act produced no leading-edge fab, and the Chips Act 2.0 proposal adopted on 3 June 2026 widens state aid and shifts the emphasis from supply to demand without changing that [43]. Japan chose a national champion instead: Rapidus verified 2 nm gate-all-around transistors on its Chitose pilot line in 2025 and raised another 267.6 billion yen, about $1.7 billion, targeting mass production in 2027 [44].

China spends more, and differently. The latest round of its National Integrated Circuit Industry Investment Fund, the Big Fund, reached $47 billion [45]. CSIS puts cumulative state funding since 2014 at $150 billion, roughly triple the CHIPS authorisation [7]. The money has produced a domestic tool industry rather than a leading-edge node.

CHIPS Act direct funding awarded, by firm, as of July 2025 $B



Source: GAO-26-107882, Semiconductors: Information on Projects Funded to Strengthen U.S. Supply Chain

Taiwan

Taiwan is the concentration risk that no policy has reduced. Its foundries hold 78 percent of the world's contract chipmaking and 92 percent of capacity for the most advanced chips, and 2 nm runs nowhere else [5]. The packaging is on the island too. Asked in October 2025 how much CoWoS capacity he would add, TSMC's chairman said only that it was "working very hard to narrow the gap between the demand and supply", with wafer and packaging capacity both "very tight" [46].

The silicon shield argument holds that this concentration deters invasion: destroying TSMC would cost an aggressor more than the island is worth. The counter is that staying irreplaceable gives Taipei reason to slow the diversification Washington wants. Taiwan bars the overseas production of its most advanced chips, so what TSMC builds in the United States runs a generation behind what it builds at home, and nobody expects Arizona to reach 2 nm before 2028 [5].

Arizona narrows the gap in wafers, and TSMC has two advanced packaging fabs planned there [46]. But a fab without its Taiwanese mask shops, chemical suppliers and trained crews is not a substitute. We think the shield is real and weakening, and that diversification is not keeping up.

Where the chips go: the Gulf, Southeast Asia, and smuggling

The Gulf bought its way inside the fence. In November 2025 Commerce authorised G42 of the United Arab Emirates and HUMAIN of Saudi Arabia to buy up to 35,000 Nvidia GB300s each under security and reporting conditions [47]. On 10 July 2026 it moved the Emirates from Country Groups D:3 and D:4 into A:5, the tier that holds Washington's close allies, so controlled AI chips and systems now ship licence-free, with no quantity cap, to the Emirati government and a named list including G42 and the US hyperscalers [48]. Saudi Arabia still works licence by licence.

Southeast Asia is where the controls leak. Since 14 July 2025 Malaysia has required a permit to export, tranship or transit any high-performance US-origin AI chip, plus 30 days' notice under its Strategic Trade Act [49]. On 25 March 2026 the Justice Department charged three people over about $170 million of servers bought through Thai front companies for delivery to China [50].

The House select committee on China puts the median at 140,000 chips smuggled to Chinese entities in 2024 [51]. Congress's answer, the Chip Security

Act, would require location verification on controlled chips within 180 days of enactment [52]. The House Foreign Affairs Committee passed it 42-0 on 26 March 2026 [53]; it has not become law.

Predicted winners

These are our judgments, not any source's.

- **Taiwan and Korea.** Nothing in the policy mix moves leading-edge logic, memory stacking or advanced packaging off an island and a peninsula before 2030; reshoring adds capacity elsewhere without subtracting it there.
- **The Netherlands and Japan.** Their tool makers are the fence, so they gain a say in who gets what and lose the business that paid for it: Tokyo Electron's China share fell from 47.4 to 26.8 percent of quarterly sales in two years [54].
- **The United States.** It holds 93 percent of accelerator design revenue and will keep it. It has built no wafer, memory or assembly industry, and $30.9 billion of grants will not.
- **China.** Mature nodes, sub-leading-edge tools, DRAM and NAND are going domestic on a visible timetable. Extreme-ultraviolet machines, their resist and high-bandwidth memory at scale are not, and stockpile arithmetic caps Ascend output near a million units a year.

The Gulf comes out ahead without making anything: Emirati entities buy advanced compute licence-free while European ones do not. The small yard is the clearest casualty: tariffs, equity stakes, a cut of licensed sales and a licence queue add up to industrial policy rather than a narrow security perimeter.

What to watch

- **November 2026.** Whether China renews the one-year suspension of its gallium ban or lets it expire [34].
- **The MATCH Act's 150-day clock.** If it passes, allied divergence on deep-ultraviolet tools gets a statutory deadline [29].
- **CXMT's memory ramp.** Domestic memory rather than logic wafers caps Chinese accelerator volume [37].
- **SMIC's advanced-node ramp.** 45,000 wafers a month at end-2025, an estimated 60,000 in 2026 and 80,000 in 2027, every one of them multiply exposed for want of an extreme-ultraviolet machine [37]. Yield is the open question.

- **Taiwan's overseas-production ban against its $250 billion investment promise.** The two contradict each other, and one will be dropped.

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Made by **Yashvardhan Sharma** with Claude. Inspired by Humanity's Last Machine and CSET's supply-chain work.

UPDATED 10 SEPTEMBER 2026

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